Western MA Per Pupil Expenditure Report

With selected districts for comparison

Table of Contents

Appendices (online)

Appendices A-D are available online at:

https://github.com/timshores/schools/blob/main/ppe_report/output/WMPPE%20Appendices.pdf

Executive Summary

"What gets measured gets managed — even when it's pointless to measure and manage it, and even if it harms the purpose of the organisation to do so"
— V. F. Ridgway, 'Dysfunctional Consequences of Performance Measurements', 1956

"By the time it can be captured in numbers, it's too late."
— Peter Drucker, 'The Effective Executive', 1967

How to use this report

The report is organized from broad regional patterns down to district-level details. As a descriptive report, it makes no forecasts and offers no recommendations. One way to use it would be as a companion to discussions that depend on meaningful cost comparisons. You could read straight through to build understanding from high-level trends to selected district details. Or you could use this **quick-start guide** to answer specific questions:

• How does my school district spending compare to Western MA baselines?
• How does Chapter 70 aid compare?
• How does required and actual net school spending compare?
• How have these categories of spending and aid grown over time in comparison with Western MA baselines?

Quick-start guide:

1. Review the cohort comparison tables at the end of the Executive Summary.
2. Review the **Scatterplot of enrollment vs. per-pupil expenditure** in Section 1 to see how all districts are grouped.
3. Locate your district in **Section 3** for detailed comparison of expenses, enrollment, Chapter 70 aid, and net school spending compared to the cohort.

Draft Status: As of **October 29, 2025**, this report is under review and has not been presented to any committee. Questions and comments should be directed to the author, Tim Shores (Leverett and Amherst-Pelham Regional School Committee member) at shorest@arps.org.

Understanding Per-Pupil Expenditure in Context

Per-pupil expenditure (PPE) provides a flawed lens for understanding school budgets. While this single metric obscures the complexity of school systems — shaped by student needs, facility requirements, governance structures, and historical commitments — it remains an accessible standard for state reporting and public discussion. We might as well try to understand it in full. This report examines PPE patterns across Western Massachusetts districts, organizing them into enrollment-based cohorts to identify trends and provide context for budget deliberations.

The Challenge of Interpretation

In February 2025, an Amherst Town Councilor asked why regional school spending was "increasing faster than the town's revenue, especially given the significant long-term reduction in the number of students we teach."[1] This question reflects a common assumption or hope that school budgets should track with municipal revenue and scale proportionally with enrollment. However, evidence of school budget trends suggests different dynamics at work.

Schools are not like factories, nonprofit organizations or hospitals. Schools are like schools, and we expect to fund and operate them like schools. As a social service, education is fundamentally labor-intensive. National data shows school districts dedicate 80% of budgets to staff and benefits — salaries, health insurance, and retirement contributions — double the 35-40% typical in other organizations.[2] These costs respond to regional and sectoral labor markets rather than local tax revenues, with the cost of education service historically rising faster than general inflation.[3] Fixed and semi-fixed costs — staffing minimums, transportation contracts, facility operations, utilities, debt service, insurance, legal — cannot be reduced proportionally with enrollment. Students eligible for special education have a right to services regardless of overall enrollment, with costs driven by individual needs rather than district size. Charter school transfers result in a net loss of state aid to tuition payments and often concentrate higher-need student populations in the traditional schools.

Approach

This report takes the position that before we set out to understand why we spend so much on schools, we should ask two key questions: What has our budget growth rate been, and what was our starting point? Understanding whether education costs have grown predictably or exceptionally is a question we can answer by looking at data collected by the Commonwealth. This report breaks down PPE and compares expenses by category, state aid, and local contributions across enrollment-based district cohorts. These groupings allow comparison of districts facing similar scale challenges. The goal is to support informed deliberation about education funding within the broader context of regional economic conditions and policy decisions.

Key Finding: Consistent Growth from Elevated Baselines

Analysis of spending patterns in Amherst-Pelham, Amherst, Leverett, Pelham, and Shutesbury reveals an important pattern: per-pupil expenditure in these districts has grown at rates comparable to — and often lower than — their Western Massachusetts peers over the 2009-2024 period. Their current high PPE reflects normal growth from a high starting point. In 2009, the selected districts already maintained PPE significantly above regional averages. With normal growth rates applied to above-average baselines, spending will remain above average.

If we learn that school budgets grow consistently at 4-6%, then we should plan for 4-6% school budget growth. Less than normal growth risks the adequacy of our schools.

Implications

This finding should shift our focus from addressing perceived runaway costs to addressing the change in local fiscal effort to fund our schools. Fiscal effort measures a community's local tax burden for schools relative to its capacity to pay. The Commonwealth calculates this as each district's required local contribution divided by its combined property values and resident income — a community spending 5% of its wealth shows lower effort than one spending 10%, regardless of absolute dollar amounts.

Two districts with identical PPE amounts and growth may differ significantly in their tax bases. This difference will show up in fiscal effort calculations. This is an objective measure of property value and income used to calculate state aid, but effort is also subjective. If fiscal effort becomes too much for local taxpayers, communities face a hard choice. In this scenario, school committees must identify educational needs and purposeful spending within available resources, but their principal concern will remain with the best interests of the district and the students. Leadership on deciding available resources falls to appropriating bodies and, ultimately, to voters.

In the end, we are likely to find that our structural funding challenges will persist until the Commonwealth finds the courage to deliver policy changes that no region, town or school district can achieve alone.

For a good start on how to advocate for meaningful change, see the July 2022 report A Sustainable Future for Rural Schools, the steadfast work of the champions at Rural Schools MA, and the September 2025 draft report by the Amherst-Pelham Regional School Committee's Fiscal Sustainability Subcommittee.[4]

References

1. Amherst Indy, "Letter: Correcting Councilor Ryan on the Regional School Budget"
https://www.amherstindy.org/2025/02/07/letter-correcting-councilor-ryan-on-the-regional-school-budget/

2. American Association of School Administrators, "School Budgets 101"
https://www.aasa.org/docs/default-source/resources/reports/school-budgets-101.pdf

3. Where's the Money Gone? Changes in the Level and Composition of Education Spending (1995) and Inflation and the Measurement of School Spending (1996)
https://www.epi.org/publication/books_wheremoneygone/
https://nces.ed.gov/pubs97/97535/97535jx1.asp

4. Rep. Blais - Rural Schools, Rural Schools Advocacy in Massachusetts, and RSC FSS Draft Report.
https://www.repblais.org/ruralschools
https://www.ruralschoolsma.org/
https://go.boarddocs.com/ma/arps/Board.nsf/goto?open&id;=DLM3SC083619

All data in this report comes from the Massachusetts Department of Elementary and Secondary Education (DESE). See Appendix A for detailed source information (https://github.com/timshores/schools/blob/main/ppe_report/output/WMPPE%20Appendices.pdf).

Executive Summary (continued)

Total PPE comparison: Western MA enrollment cohorts and selected districts

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp	Above baseline
	Baseline
CAGR = (End/Start)^(1/years) – 1	Below baseline

Table 1

Cohort/District	2009 $/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 $/pupil
Western MA (all, excl. Springfield)	$12,234	+4.0%	+4.6%	+6.0%	$22,065
Western MA Tiny (0-200 FTE)	$14,738	+4.2%	+4.4%	+6.0%	$27,504
Western MA Small (201-800 FTE)	$12,523	+4.6%	+5.1%	+6.7%	$24,622
Western MA Medium (801-1600 FTE)	$13,032	+4.1%	+4.7%	+6.1%	$23,785
Western MA Large (1601-10K FTE)	$11,775	+3.9%	+4.5%	+5.9%	$20,841
Outliers (Springfield)	$14,608	+4.1%	+5.7%	+9.4%	$26,615

Table 2

Cohort/District	2009 $/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 $/pupil
Western MA Medium (801-1600 FTE)	$13,032	+4.1%	+4.7%	+6.1%	$23,785
Amherst-Pelham Regional	$16,211	+3.8%	+3.8%	+4.8%	$28,233
Amherst	$16,029	+4.6%	+4.7%	+5.7%	$31,267

Table 3

Cohort/District	2009 $/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 $/pupil
Western MA Tiny (0-200 FTE)	$14,738	+4.2%	+4.4%	+6.0%	$27,504
Leverett	$15,156	+2.5%	+0.5%	+2.0%	$21,910
Pelham	$14,733	+3.0%	+3.3%	+5.9%	$23,040
Shutesbury	$14,011	+4.4%	+5.3%	+5.0%	$26,908

Table 1 compares PPE and cost growth of Western MA school district cohorts with the benchmark of all Western MA districts. **Tables 2 and 3** benchmark districts against their cohort averages.

Cohorts are organized by enrollment size. Grouping districts by enrollment size enables more meaningful cost comparisons—districts of similar size face similar challenges with administration, staffing, facilities, and program requirements. Detailed information about cohorts follow in **sections 1 and 2**. Detailed comparisons of individual districts to cohorts follow in **sections 2 and 3**.

Why the CAGR threshold is more sensitive

Small differences in growth rates compound over time. A 0.5pp difference may seem modest, but:

- $20,000 PPE growing at 4.0% → $36,019 after 15 years
- $20,000 PPE growing at 4.5% → $38,706 after 15 years
- Gap: $2,687/pupil (7.5% more total growth)

For a 100-student district, that's $268,700 in additional annual spending by year 15. The 0.5pp threshold helps identify districts on meaningfully different long-term trajectories.

Executive Summary (continued)

Chapter 70 Aid (per foundation pupil): Western MA enrollment cohorts and selected districts

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp	Above baseline
	Baseline
CAGR = (End/Start)^(1/years) – 1	Below baseline

Table 4

Cohort/District	2009 $/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 $/pupil
Western MA (all, excl. Springfield)	$5,444	+3.6%	+4.4%	+6.6%	$9,313
Western MA Tiny (0-200 FTE)	$3,297	+2.2%	+2.8%	+5.0%	$4,547
Western MA Small (201-800 FTE)	$3,672	+2.7%	+3.1%	+3.8%	$5,465
Western MA Medium (801-1600 FTE)	$5,698	+3.2%	+4.0%	+5.8%	$9,202
Western MA Large (1601-10K FTE)	$5,742	+3.8%	+4.6%	+7.0%	$10,065
Outliers (Springfield)	$10,452	+4.5%	+5.6%	+8.1%	$20,138

Table 5

Cohort/District	2009 $/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 $/pupil
Western MA Medium (801-1600 FTE)	$5,698	+3.2%	+4.0%	+5.8%	$9,202
Amherst-Pelham Regional	$5,773	+2.3%	+2.5%	+2.5%	$8,068
Amherst	$4,518	+2.2%	+2.4%	+3.1%	$6,271

Table 6

Cohort/District	2009 $/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 $/pupil
Western MA Tiny (0-200 FTE)	$3,297	+2.2%	+2.8%	+5.0%	$4,547
Leverett	$1,782	+3.1%	+2.3%	+5.5%	$2,798
Pelham	$1,983	-0.1%	+1.4%	+2.5%	$1,963
Shutesbury	$3,808	+2.9%	+4.6%	+3.5%	$5,862

Tables 4-6 compare Chapter 70 aid (per foundation pupil) across Western MA school districts organized by enrollment size.

Chapter 70 aid and net school spending calculations (this and next page) use foundation enrollment for per-pupil figures, while PPE totals and expense categories (previous page) use in-district FTE enrollment. These metrics come from different DESE departments:

• **Foundation enrollment** (Chapter 70): All students residing in the district on October 1st of the prior year for whom the district is financially responsible.
• **In-district FTE** (PPE reports): Full-year enrollment calculated from total membership days for students either residing in the district, or choiced or tuitioned into the district.

This difference complicates direct comparison between Chapter 70 and expense analyses, but remains valid since each district uses the same denominator type within each metric. For more information, see the DESE Researcher's Guide.
https://www.doe.mass.edu/research/researchers.html

Executive Summary (continued)

Actual NSS above Required NSS (per foundation pupil): Western MA enrollment cohorts and selected districts

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp	Above baseline
	Baseline
CAGR = (End/Start)^(1/years) – 1	Below baseline

Table 7

Cohort/District	2009 $/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 $/pupil
Western MA (all, excl. Springfield)	$1,394	+5.9%	+8.0%	+2.6%	$3,301
Western MA Tiny (0-200 FTE)	$4,523	+5.6%	+6.3%	+4.1%	$10,285
Western MA Small (201-800 FTE)	$2,054	+7.0%	+9.1%	+5.9%	$5,656
Western MA Medium (801-1600 FTE)	$2,318	+4.2%	+8.8%	+3.3%	$4,315
Western MA Large (1601-10K FTE)	$934	+6.7%	+7.7%	+1.4%	$2,455
Outliers (Springfield)	-$169	-59.5%	-52.4%	+24.0%	-$0

Table 8

Cohort/District	2009 $/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 $/pupil
Western MA Medium (801-1600 FTE)	$2,318	+4.2%	+8.8%	+3.3%	$4,315
Amherst-Pelham Regional	$3,584	+3.9%	+4.5%	+2.1%	$6,383
Amherst	$6,217	+4.3%	+5.8%	+3.9%	$11,670

Table 9

Cohort/District	2009 $/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 $/pupil
Western MA Tiny (0-200 FTE)	$4,523	+5.6%	+6.3%	+4.1%	$10,285
Leverett	$5,429	+2.4%	+0.6%	-1.9%	$7,799
Pelham	$5,561	+4.0%	+6.6%	+7.9%	$10,086
Shutesbury	$4,173	+6.4%	+5.7%	+2.5%	$10,526

Tables 7-9 compare Actual NSS above Required NSS (per foundation pupil) - showing local funding effort beyond state requirements.

Section 1 — Western MA traditional public school district trends

Section 1 examines per-pupil expenditure (PPE) trends across all Western Massachusetts traditional school districts, organized by enrollment-based cohorts.

This section provides a regional overview before diving into cohort-specific details (Section 2) and individual district comparisons (Section 3).

Contents

- **Per-pupil expenditure and growth 2009-2024** compares 2009 to 2024 PPE for Western MA districts.
- **Year-over-Year (YoY) growth rates** shows the rate of PPE change from 2009-2024 for Amherst-Pelham, Amherst, Leverett, Pelham, Shutesbury, and their cohorts.
- **5-year and 15-year CAGR by district and cohort** shows how compound annual growth rate (CAGR) smooths out the volatility of YoY rates of change, making it easier to understand and compare change in PPE over time.
- **Distribution of 2024 enrollment and proposed cohort grouping** shows a histogram of Western MA districts by enrollment with cohorts determined by statistics visualized by the histogram.
- **Scatterplot analysis** shows the relationship between district enrollment and PPE for 2024, with a table of cohort members.
- **Maps** of district locations and 2024 enrollment cohorts, 2024 PPE vs enrollment cohort baseline, and PPE growth (2009-2024) vs enrollment cohort baseline.
- **Statistical associations** between enrollment and PPE.

All districts are grouped into enrollment cohorts based on 2024 in-district FTE using IQR (Interquartile Range) analysis. This grouping enables meaningful peer comparisons while acknowledging that enrollment size is just one factor influencing costs. Districts in the same cohort can differ significantly in wealth, demographics, geography, facility age, and governance—all of which shape costs and outcomes.

Report Navigation:

Section 1 — Western MA traditional public school district trends

Per-pupil expenditure and growth: 2009 PPE (lighter) to 2024 PPE (darker segment)

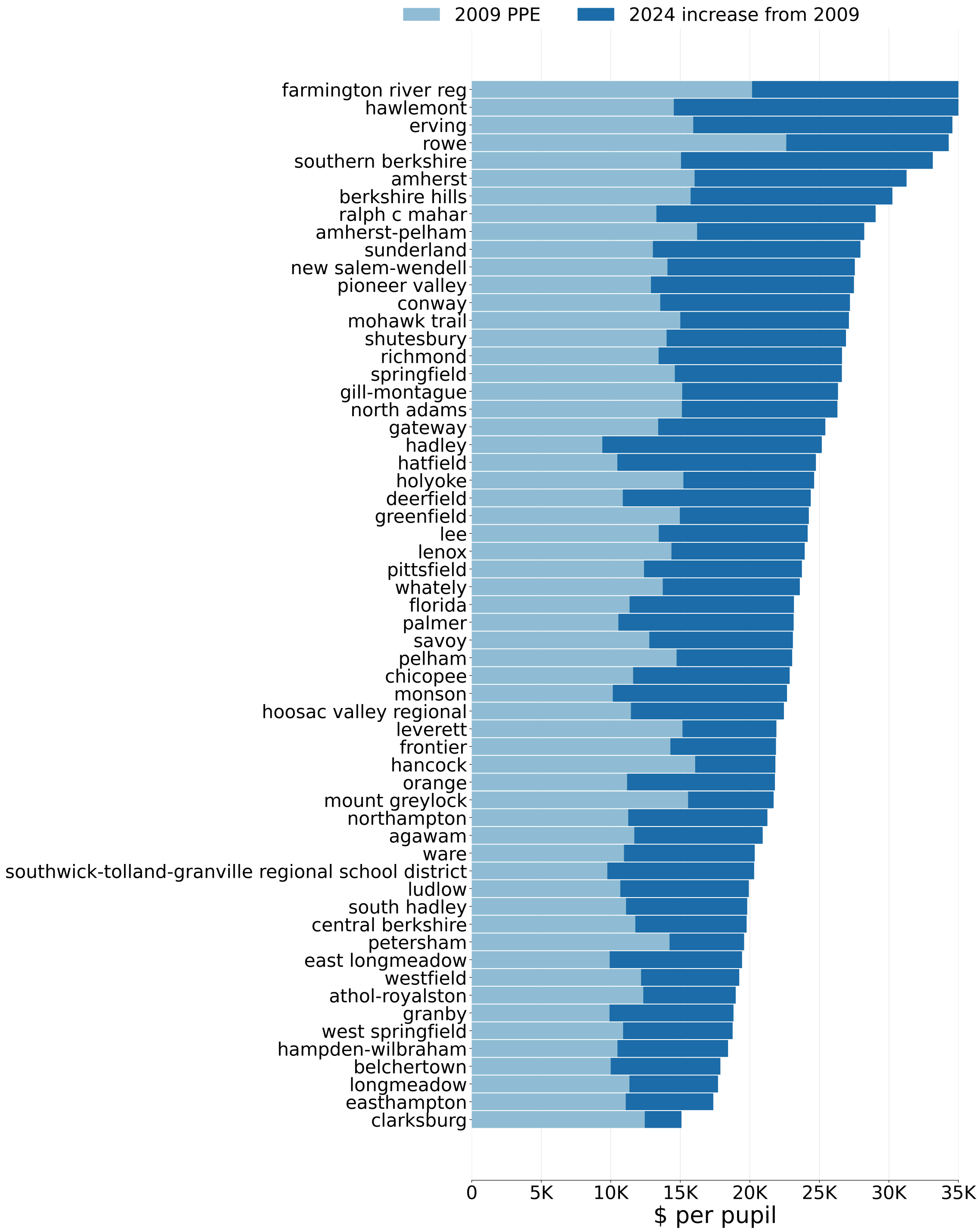


Figure 1

Each bar represents a district's per-pupil expenditures (PPE) in 2024, with growth from 2009 shown by a darker color. Districts are sorted by 2024 spending. While the spread is visible, this alone is nearly meaningless: we don't know to what degree the difference between $15K and $30K per student reflects demographic change, education quality, cost-of-living adjustments, special education demand, economies of scale, administrative decisions, ballot box decisions, or legacy cost structures. The next page zeroes in on rates of change in selected districts.

Note: The following districts are omitted from this analysis: Chesterfield-Goshen (missing expenditure data), Hampshire (missing expenditure data), Southampton (missing expenditure data), Warwick (no enrollment data), Westhampton (missing expenditure data), Williamsburg (missing expenditure data).

Section 1 — Western MA traditional public school district trends

Year-over-Year (YoY) growth rates by district and cohort

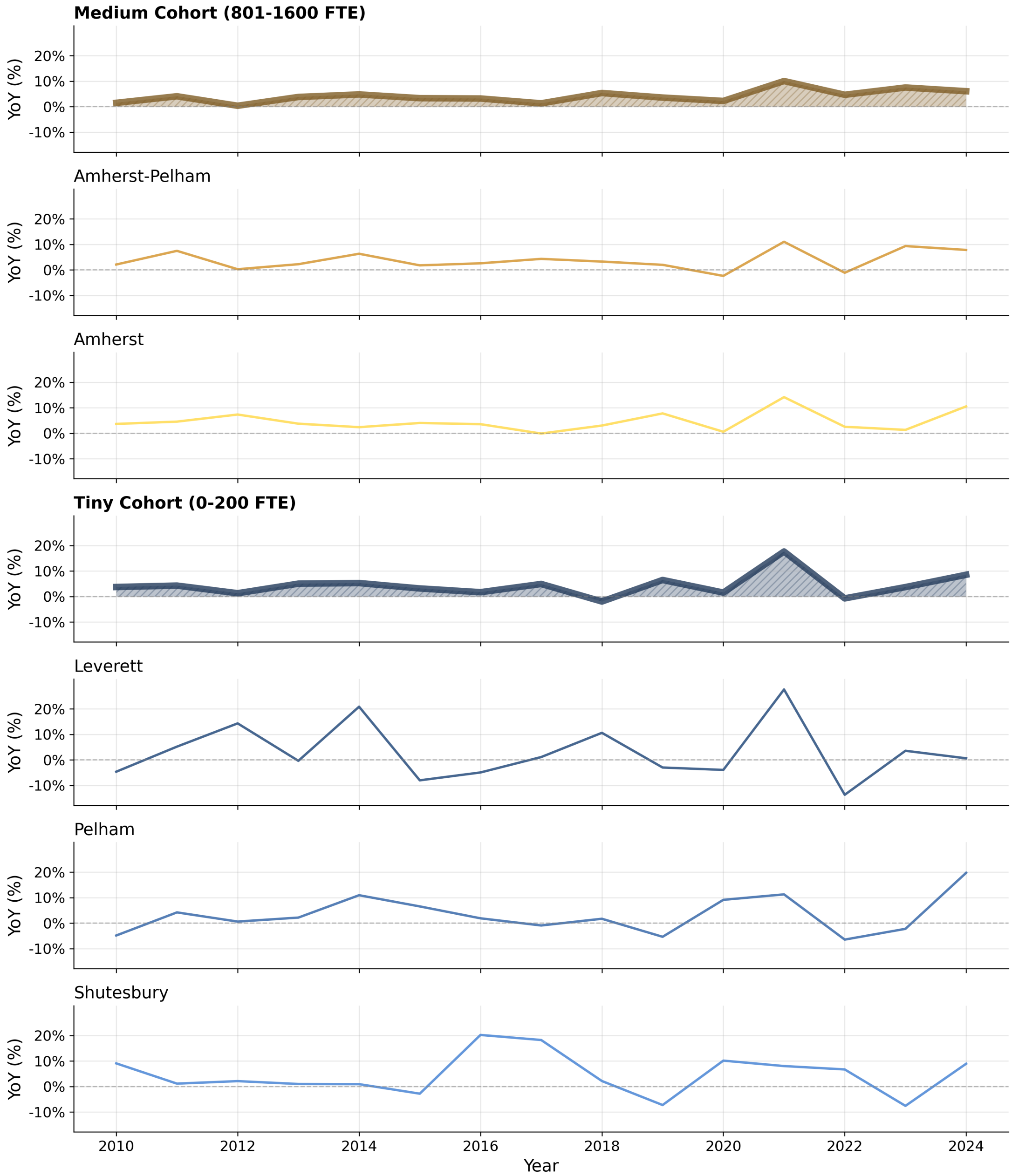


Figure 2

These plots show year-over-year (YoY) growth rates in per-pupil expenditure (PPE) for districts of interest and their enrollment cohorts. Thicker lines represent cohort aggregates, thinner lines show individual districts. Cohort determination is detailed in Appendix A.[1] While patterns emerge, no clear, actionable signal appears across all districts and cohorts over the 2009-2024 period. On the next page, we'll see how a compound growth calculation makes it easier to see similarities and differences between districts.

1. https://github.com/timshores/schools/blob/main/ppe_report/output/WMPPE%20Appendices.pdf

Section 1 — Western MA traditional public school district trends

5-year and 15-year CAGR by district and cohort

Compound annual growth rates (CAGR) smooth out YoY volatility to reveal long-term trends. Below, districts are color-coded by enrollment cohort (blues for Tiny, golds for Medium), with white diagonal lines marking cohort baselines. Comparing three 5-year periods (2009-2014, 2014-2019, 2019-2024) shows how growth rates have shifted over time. The second chart displays overall 15-year CAGR from 2009 to 2024.

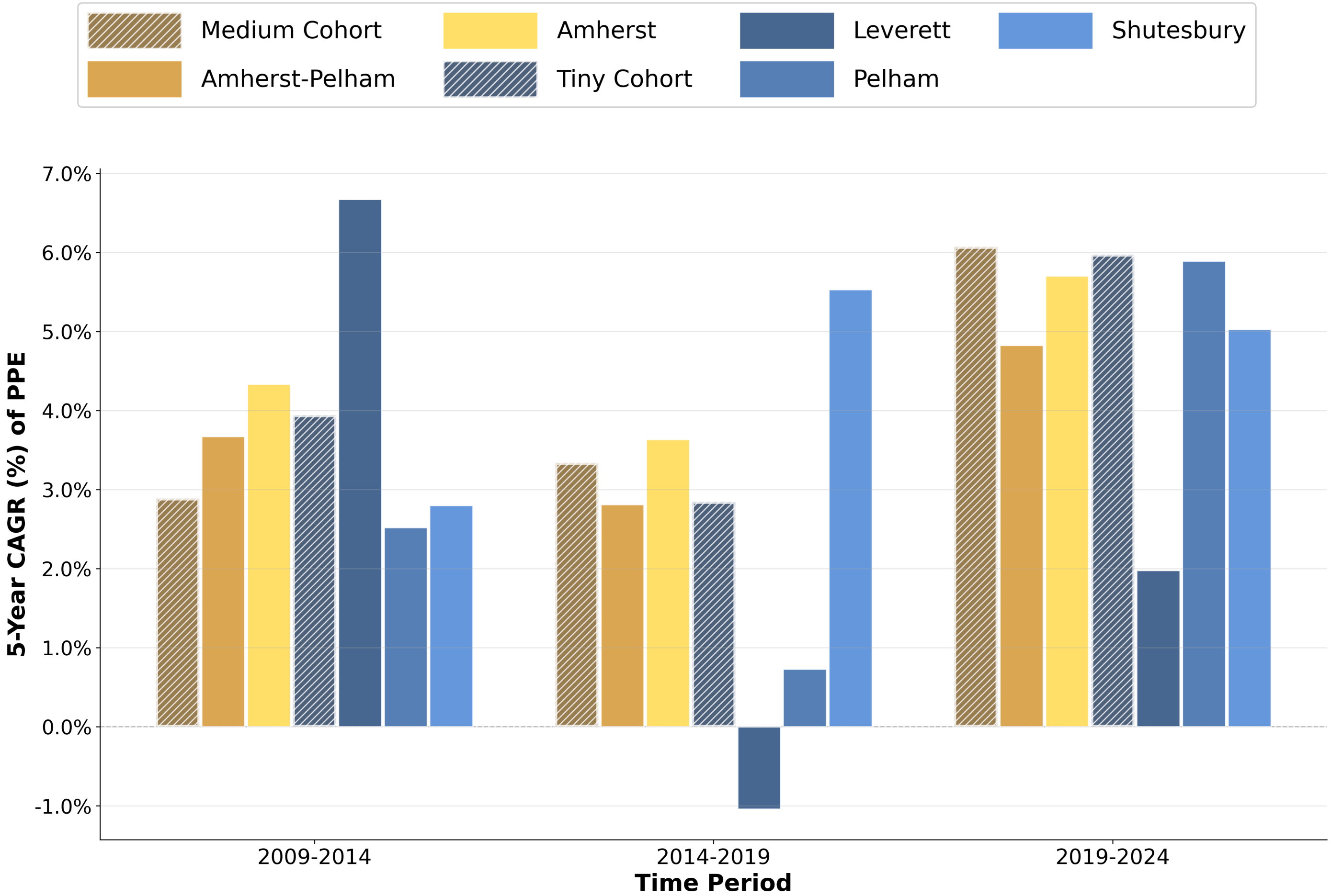


Figure 3

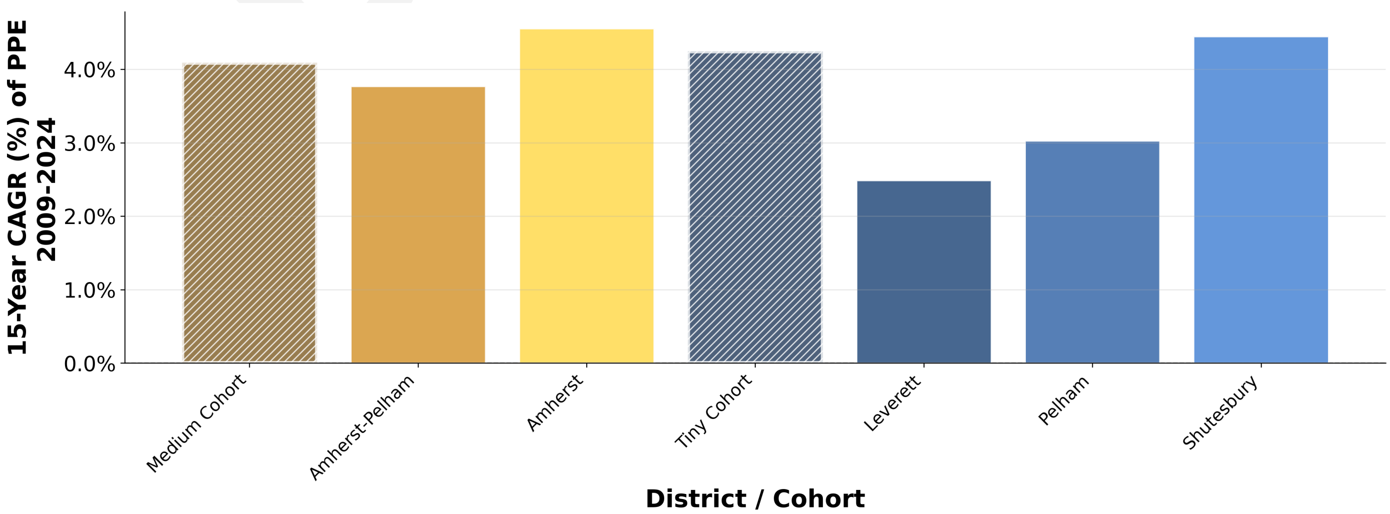


Figure 4

Section 1 — Western MA traditional public school district trends

Distribution of 2024 enrollment and proposed cohort grouping

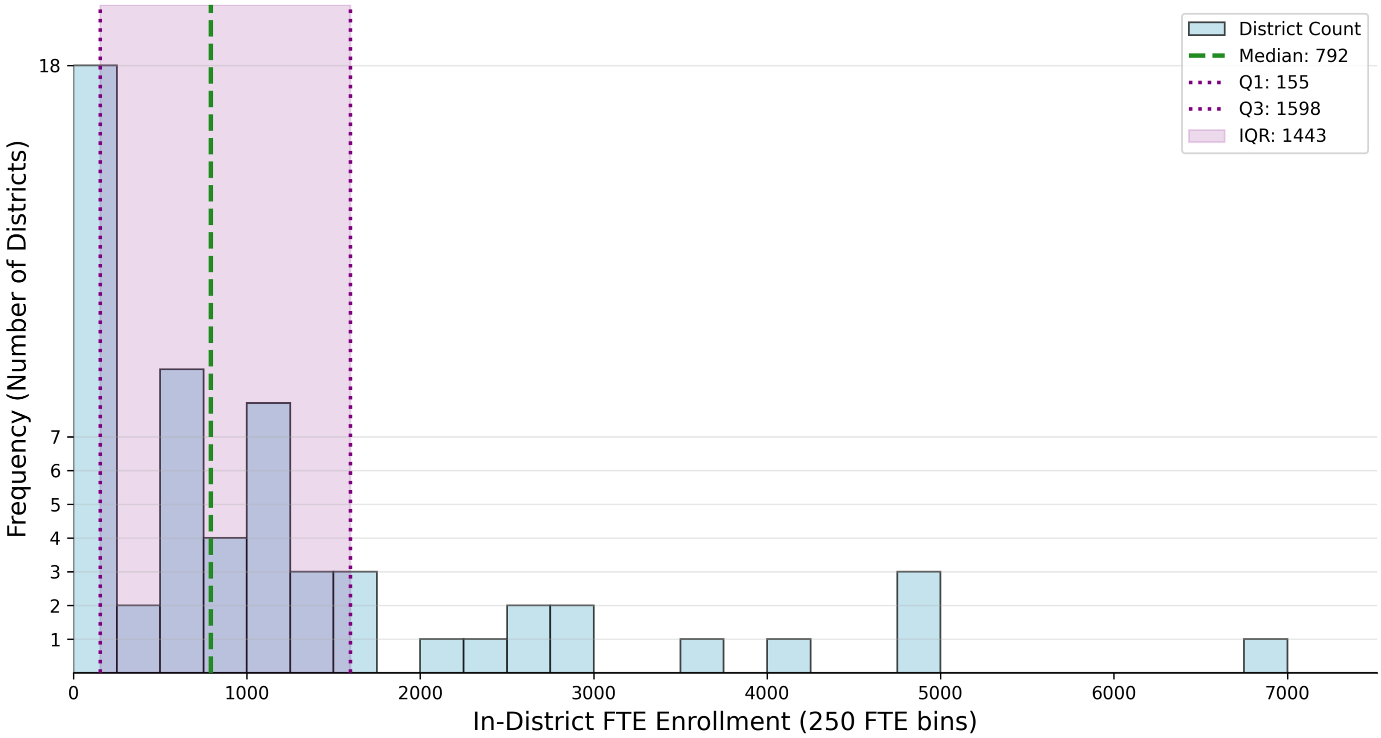


Figure 5

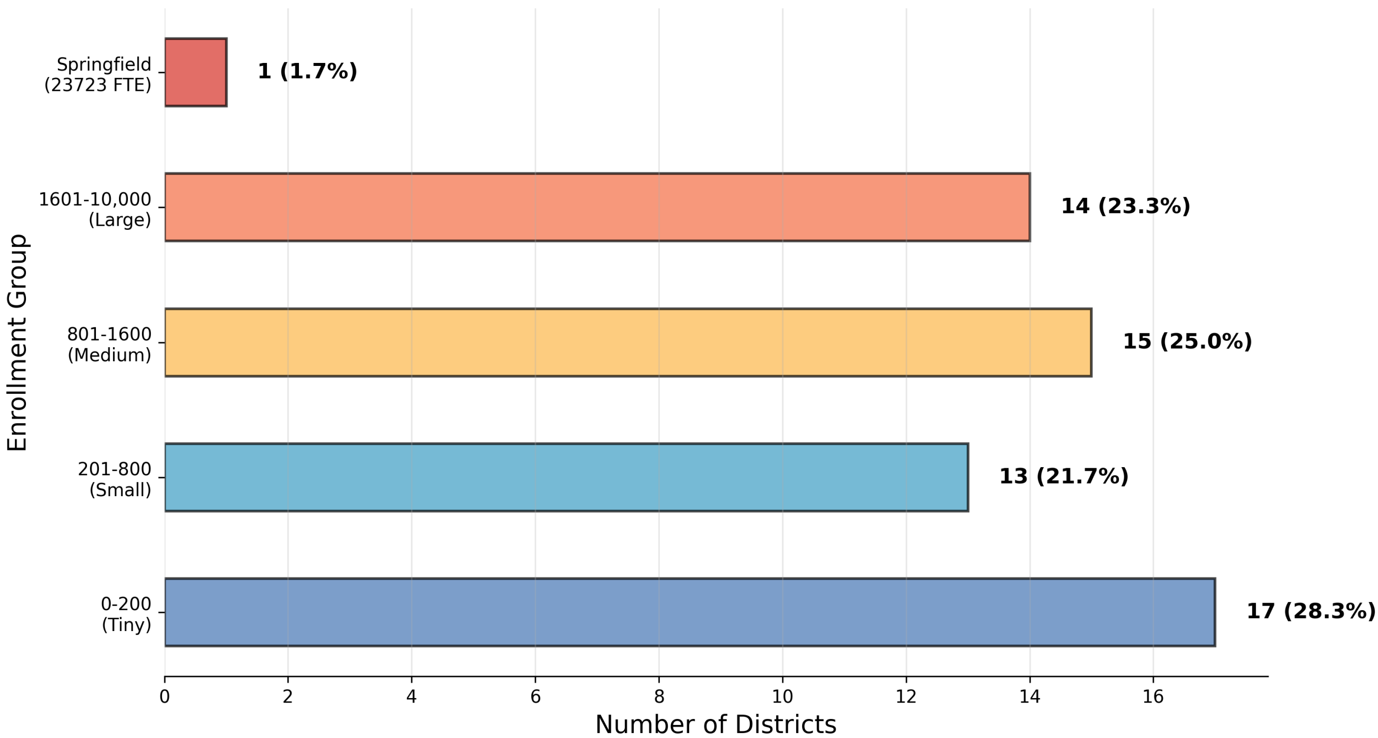


Figure 6

District enrollment cohorts need clear boundaries. To avoid arbitrary cutoffs, this approach uses the natural spread of enrollments. The histogram (top) shows a right-skewed distribution of 2024 enrollments with a long tail of larger districts. Based on interquartile range analysis, districts are grouped into four cohorts (bottom): Tiny (0-200 FTE), Small (201-800 FTE), Medium (801-1600 FTE), and Large (1601-10,000 FTE). Springfield, with an in-district enrollment of 23,723 in 2024, is omitted as an outlier. These cohorts enable comparison by scale to discover financial norms, but are not meant to imply what spending should or should not be for schools of different sizes.

The next page presents each district's enrollment and PPE in both scatterplot and reference table format.

Section 1 — Western MA traditional public school district trends

Scatterplot of enrollment vs. per-pupil expenditure with quartile boundaries (2024)

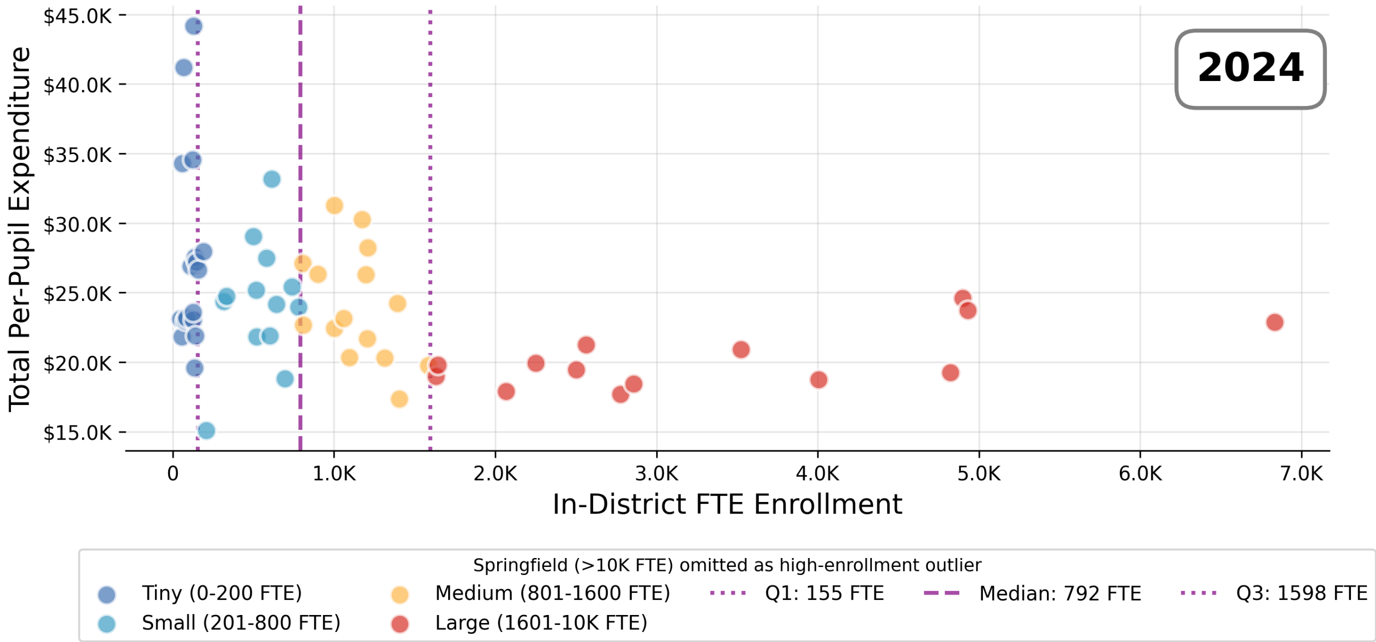


Figure 7

Table 10

District	2024 In-District FTE	2024 PPE ▼
Farmington River Reg (Regional K-12)	128	$44,178
Hawlemont (Regional K-12)	67	$41,189
Erving	123	$34,566
Rowe	60	$34,297
Sunderland	191	$27,953
New Salem-Wendell	138	$27,553
Conway	146	$27,200
Shutesbury	110	$26,908
Richmond	158	$26,631
Whately	127	$23,593
Florida	86	$23,175
Savoy	47	$23,099
Pelham	126	$23,040
Worthington	77	$22,985
Leverett	142	$21,910
Hancock	56	$21,836
Petersham	134	$19,591
Southern Berkshire (Regional K-12)	614	$33,162
Ralph C Mahar (Secondary Region)	500	$29,040
Pioneer Valley (Regional K-12)	581	$27,490
Gateway (Regional K-12)	740	$25,430
Hadley	516	$25,173
Hatfield	332	$24,752
Deerfield	317	$24,382
Lee	643	$24,155
Lenox	779	$23,944
Frontier (Secondary Region)	602	$21,881
Orange	521	$21,809
Granby	695	$18,824
Clarksburg	208	$15,078
Amherst	1,002	$31,267
Berkshire Hills	1,174	$30,257
Amherst-Pelham (Secondary Region)	1,209	$28,233
Mohawk Trail (Regional K-12)	805	$27,124
Gill-Montague (Regional K-12)	900	$26,345
North Adams	1,196	$26,300
Greenfield	1,394	$24,237
Palmer	1,062	$23,154
Monson	808	$22,675
Hoosac Valley Regional	1,003	$22,450
Mount Greylock (Regional K-12)	1,207	$21,705
Ware	1,096	$20,348
Southwick-Tolland-Granville Regional School District (Regional K-12)	1,313	$20,311
Central Berkshire (Regional K-12)	1,586	$19,773
Easthampton	1,404	$17,371
Holyoke	4,899	$24,619
Pittsfield	4,931	$23,739
Chicopee	6,834	$22,866
Northampton	2,562	$21,267
Agawam	3,523	$20,920
Ludlow	2,251	$19,928
South Hadley	1,643	$19,810
East Longmeadow	2,503	$19,444
Westfield	4,821	$19,236
Athol-Royalston	1,633	$18,981
West Springfield	4,004	$18,760
Hampden-Wilbraham (Regional K-12)	2,858	$18,436
Belchertown	2,068	$17,887
Longmeadow	2,777	$17,710
Springfield	23,723	$26,615

Section 1 — Western MA traditional public school district trends

Geographic map showing district locations and enrollment cohorts (2024)

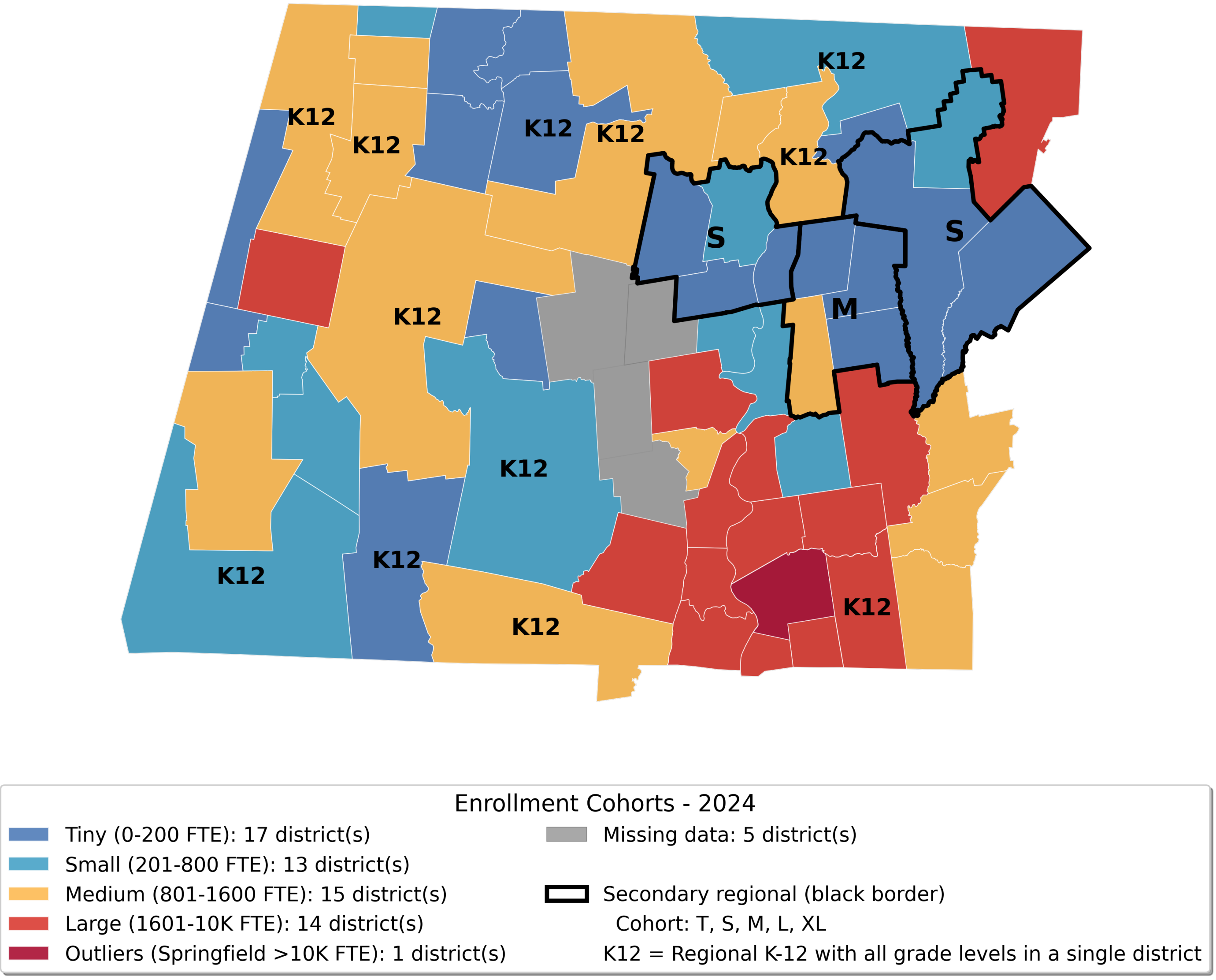


Figure 8

Geography matters for understanding school costs. This map shows where enrollment cohorts cluster and how regional district boundaries cross municipal lines.

Single-town elementary and PK-12 regions appear as solid shaded areas, colored by their enrollment cohort (blue for Tiny, light blue for Small, yellow for Medium, orange for Large). Regional K-12 districts (marked 'K12') span multiple towns. Secondary regional districts are outlined with thick black borders and labeled (T/S/M/L) to show their cohort without covering up their member elementary districts.

See also the [MassGIS map of Public School Districts.](https://massgis.maps.arcgis.com/apps/webappviewer/index.html?id=5b652451857f41ecabc5aa96ef53641c)
https://massgis.maps.arcgis.com/apps/webappviewer/index.html?id=5b652451857f41ecabc5aa96ef53641c

Section 1 — Western MA traditional public school district trends

Geographic map showing 2024 PPE vs enrollment cohort baseline

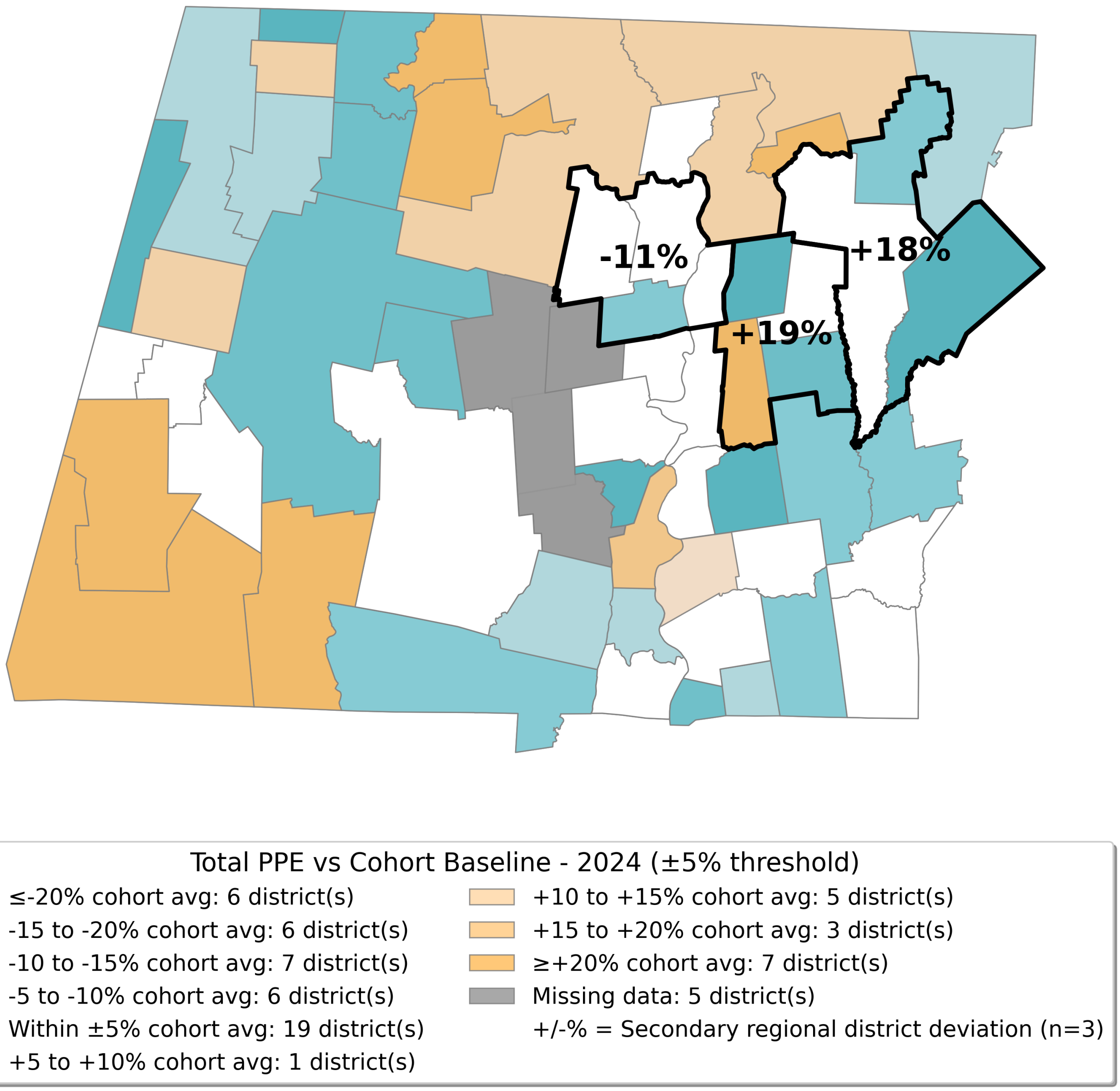


Figure 9

This map shows each district's 2024 per-pupil spending compared to its enrollment cohort baseline (weighted average of all Western MA districts in that cohort). This helps identify which districts are operating above or below the typical spending level for their enrollment size, accounting for the well-documented relationship between district size and per-pupil costs.

Color coding: Dark green indicates districts spending more than 10% below their cohort average (most cost-efficient). Light green shows 0-10% below average. Light red shows 0-10% above average. Dark red indicates districts spending more than 10% above their cohort average (highest relative cost).

See also the [MassGIS map of Public School Districts.](https://massgis.maps.arcgis.com/apps/webappviewer/index.html?id=5b652451857f41ecabc5aa96ef53641c)
https://massgis.maps.arcgis.com/apps/webappviewer/index.html?id=5b652451857f41ecabc5aa96ef53641c

Section 1 — Western MA traditional public school district trends

Geographic map showing 15-year PPE growth (2009-2024) vs enrollment cohort baseline

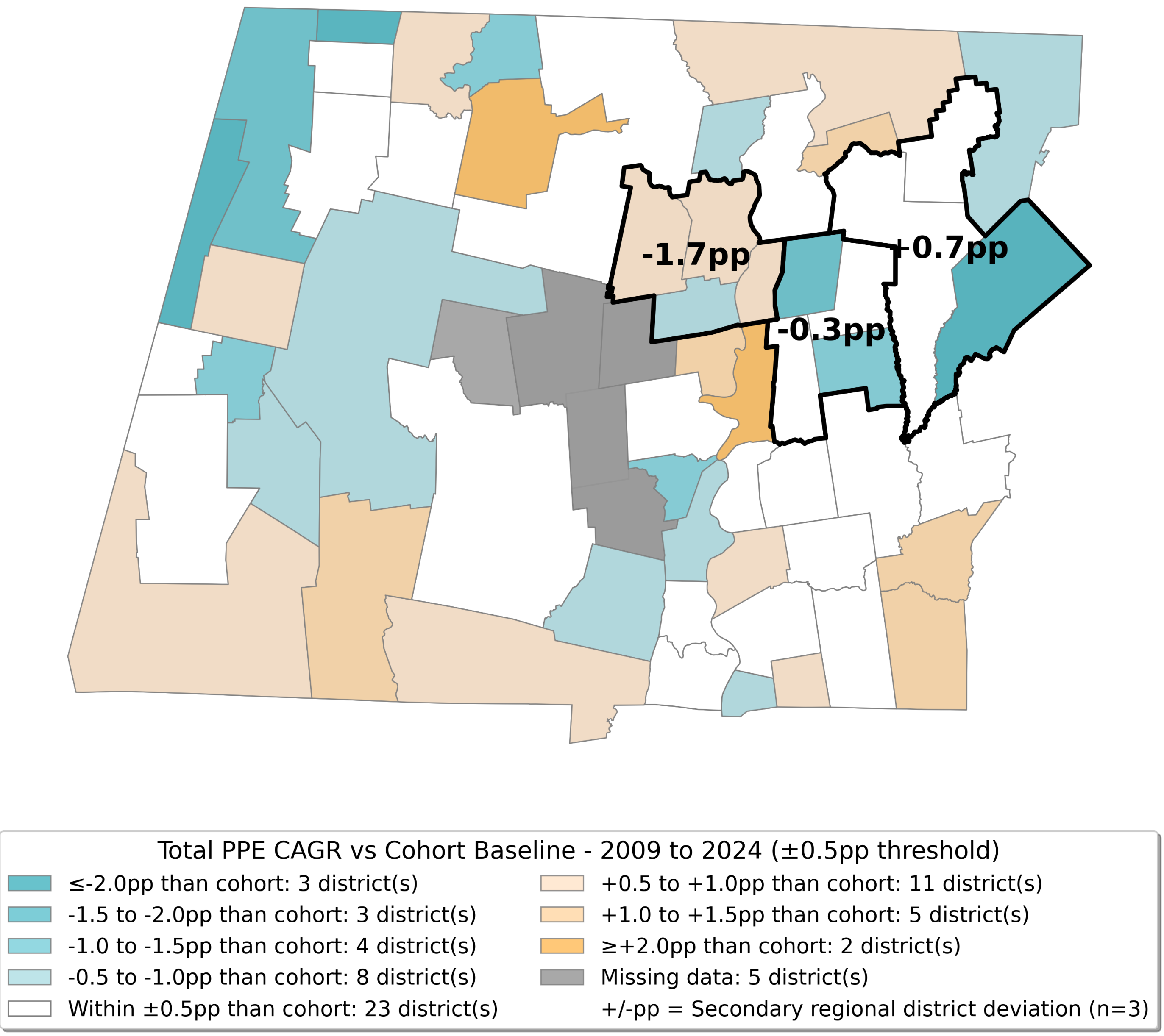


Figure 10

This map shows each district's 15-year spending growth rate (CAGR, 2009-2024) compared to its enrollment cohort baseline growth rate. This helps identify which districts experienced faster or slower spending growth than similar-sized districts, revealing local policy decisions and cost pressures beyond typical enrollment cohort trends.

Color coding: Dark blue indicates districts with growth more than 0.5 percentage points slower than their cohort average. Light blue shows 0-0.5 pp slower. Light orange shows 0-0.5 pp faster. Dark orange indicates districts with growth more than 0.5 pp faster than their cohort average.

See also the MassGIS map of Public School Districts.
https://massgis.maps.arcgis.com/apps/webappviewer/index.html?id=5b652451857f41ecabc5aa96ef53641c

Section 1 — Western MA traditional public school district trends (continued)

Statistical Associations between Enrollment and Per-Pupil Expenditures

Key Finding

District size and enrollment trends explain very little about spending patterns. Analysis of Western Massachusetts traditional districts (2009-2024) reveals that enrollment cohort (district size) explains only 22% of variation in current spending levels and has no association with spending growth rates. Enrollment growth rates show even weaker associations. This is evidence that knowing a district's size or enrollment trajectory tells us little about its per-pupil expenditure.

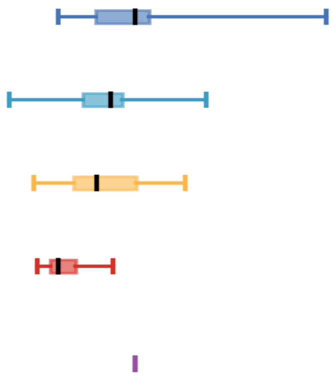
Distribution of PPE by Enrollment Cohort

The five-number summaries and box plots below show two critical dimensions: where districts are now (2024 PPE) and how they got there (2009-2024 PPE growth rates).

2024 PPE by Cohort

Box plots show the spread of current spending within each size cohort. Wide boxes indicate greater spending variation; narrow boxes suggest similar spending patterns. The median line shows typical spending for each cohort.

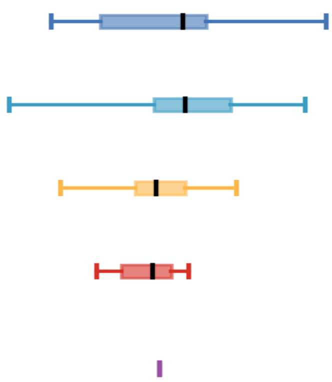
Cohort	n	Min	Q1	Median	Q3	Max	Distribution
Tiny (0-200 FTE)	17	$19,591	$23,040	$26,631	$27,953	$44,178	
Small (201-800 FTE)	13	$15,078	$21,881	$24,382	$25,430	$33,162	
Medium (801-1600 FTE)	15	$17,371	$21,026	$23,154	$26,734	$31,267	
Large (1601-10K FTE)	14	$17,710	$18,815	$19,627	$21,180	$24,619	
Outliers (Springfield >10K FTE)	1	$26,615	$26,615	$26,615	$26,615	$26,615	



2009-2024 PPE Growth Rates by Cohort

These box plots reveal whether districts of similar size grew at similar rates (narrow boxes) or diverged over time (wide boxes). Comparing medians shows which cohorts experienced faster or slower growth.

Cohort	n	Min	Q1	Median	Q3	Max	Distribution
Tiny (0-200 FTE)	16	2.1%	3.0%	4.5%	5.0%	7.2%	
Small (201-800 FTE)	13	1.3%	4.0%	4.6%	5.4%	6.8%	
Medium (801-1600 FTE)	15	2.2%	3.6%	4.0%	4.6%	5.5%	
Large (1601-10K FTE)	14	2.9%	3.4%	3.9%	4.3%	4.6%	
Outliers (Springfield >10K FTE)	1	4.1%	4.1%	4.1%	4.1%	4.1%	



Pattern Recognition: A cohort with high 2024 PPE but typical growth rates suggests historically elevated spending maintained over time — exactly what we see in our selected districts.

Section 1 — Western MA traditional public school district trends (continued)

Statistical Test Results for Per-Pupil Expenditures

Statistical Test Results for Per-Pupil Expenditures

What Associates with Current Spending (2024 PPE)?

District size (cohort) — WEAK ASSOCIATION

• ANOVA: $F(3,36) = 3.85$, $p = 0.014$, $\eta^2 = 0.216$
• Enrollment cohort explains ~22% of PPE variation
• Effect size: small to medium (substantial variation within cohorts)

Enrollment growth rate — NO ASSOCIATION

• Linear regression: $r = 0.226$, $R^2 = 0.051$, $p = 0.160$
• Explains only 5% of PPE variation
• Not statistically significant at $\alpha = 0.05$

Combined model — MODEST PREDICTIVE POWER

• ANCOVA: $R^2 = 0.232$, Adjusted $R^2 = 0.155$, $F(4,35) = 3.02$, $p = 0.023$
• Size + growth together explain 23% of variation
• Enrollment growth adds minimal predictive value once cohort is accounted for

What Associates with Spending Growth (2009-2024 CAGR)?

District size (cohort) — NO ASSOCIATION

• ANOVA: $F(3,36) = 0.61$, $p = 0.611$, $\eta^2 = 0.036$
• Effect size: negligible
• All cohorts show similar growth rate distributions

Enrollment growth rate — NO ASSOCIATION

• Linear regression: $r = -0.175$, $R^2 = 0.031$, $p = 0.283$
• Weak negative correlation (not significant)
• Declining enrollment doesn't predict higher PPE growth

Combined model — NO PREDICTIVE POWER

• ANCOVA: $R^2 = 0.069$, Adjusted $R^2 = -0.027$, $F(4,35) = 0.74$, $p = 0.569$
• Negative adjusted R^2 indicates model performs worse than sample mean
• Enrollment characteristics don't predict spending trajectories

Interpretation

These analyses examine statistical associations, not causal relationships. The weak associations demonstrate that:

• Districts of similar size have widely varying spending levels
• Enrollment change doesn't systematically drive PPE change
• Other factors — wealth, demographics, facility age, governance, historical commitments — matter more than enrollment characteristics

Section 1 — Western MA traditional public school district trends (continued)

Statistical Associations between Enrollment and Chapter 70 Aid (per foundation pupil)

Key Finding

District size and enrollment cohort show associations with Chapter 70 Aid patterns. This section examines the distribution of Chapter 70 state aid (per foundation pupil) across Western Massachusetts traditional districts organized by enrollment size.

Distribution of Chapter 70 Aid by Enrollment Cohort

The five-number summaries below show two dimensions: current aid levels (2024 Ch70 Aid) and how they grew over time (2009-2024 growth rates).

2024 Chapter 70 Aid per Foundation Pupil by Cohort

Box plots show the spread of state aid within each size cohort. The median line shows typical aid levels for each cohort. Note: Foundation enrollment includes weighted adjustments for low-income, ELL, and special education students.

Cohort	n	Min	Q1	Median	Q3	Max	Distribution
Tiny (0-200 FTE)	15	$1,963	$3,056	$4,531	$6,273	$11,113	
Small (201-800 FTE)	10	$1,727	$2,863	$4,262	$7,733	$13,208	
Medium (801-1600 FTE)	8	$6,173	$7,619	$9,099	$11,187	$12,345	
Large (1601-10K FTE)	11	$2,557	$5,371	$7,103	$12,669	$19,792	
Outliers (Springfield >10K FTE)	1	$20,138	$20,138	$20,138	$20,138	$20,138	

2009-2024 Chapter 70 Aid Growth Rates by Cohort

These box plots reveal whether districts of similar size received similar aid growth (narrow boxes) or diverged over time (wide boxes). Comparing medians shows which cohorts experienced faster or slower aid increases.

Cohort	n	Min	Q1	Median	Q3	Max	Distribution
Tiny (0-200 FTE)	15	-1.1%	0.9%	2.4%	3.6%	13.2%	
Small (201-800 FTE)	10	0.0%	1.3%	2.8%	3.3%	5.6%	
Medium (801-1600 FTE)	8	1.5%	2.2%	3.3%	4.0%	4.1%	
Large (1601-10K FTE)	11	1.1%	2.1%	3.3%	3.9%	5.2%	
Outliers (Springfield >10K FTE)	1	4.5%	4.5%	4.5%	4.5%	4.5%	

Section 1 — Western MA traditional public school district trends (continued)

Statistical Test Results for Chapter 70 Aid (per foundation pupil)

Statistical Test Results for Chapter 70 Aid

What Associates with Current Chapter 70 Aid (2024 Ch70 per foundation pupil)?

District size (cohort) — MODERATE ASSOCIATION

- ANOVA: $F(3,36) = 6.12$, $p = 0.002$, $\eta^2 = 0.338$
- Enrollment cohort explains ~34% of Ch70 Aid variation
- Effect size: medium (smaller districts receive higher per-pupil aid due to foundation formula minimum)

Enrollment growth rate — MINIMAL ASSOCIATION

- Linear regression: $r = -0.142$, $R^2 = 0.020$, $p = 0.389$
- Explains only 2% of Ch70 Aid variation
- Not statistically significant at $\alpha = 0.05$

Combined model — MODEST PREDICTIVE POWER

- ANCOVA: $R^2 = 0.351$, Adjusted $R^2 = 0.277$, $F(4,35) = 5.46$, $p = 0.002$
- Size + growth together explain 35% of variation
- Enrollment growth adds minimal predictive value once cohort is accounted for

What Associates with Chapter 70 Aid Growth (2009-2024 CAGR)?

District size (cohort) — WEAK ASSOCIATION

- ANOVA: $F(3,36) = 1.28$, $p = 0.297$, $\eta^2 = 0.096$
- Effect size: small
- All cohorts show similar aid growth rate distributions

Enrollment growth rate — NO ASSOCIATION

- Linear regression: $r = -0.084$, $R^2 = 0.007$, $p = 0.616$
- Weak negative correlation (not significant)
- Declining enrollment doesn't predict higher or lower Ch70 growth

Combined model — NO PREDICTIVE POWER

- ANCOVA: $R^2 = 0.103$, Adjusted $R^2 = 0.000$, $F(4,35) = 1.13$, $p = 0.358$
- Near-zero adjusted R^2 indicates model performs no better than sample mean
- Enrollment characteristics don't predict aid growth trajectories

Interpretation

These analyses examine statistical associations, not causal relationships. The findings demonstrate that:

- Chapter 70 Aid levels associate with district size due to foundation budget formula minimums
- Aid growth rates show no association with enrollment characteristics
- State funding policy changes and formula adjustments affect all districts similarly

Section 1 — Western MA traditional public school district trends (continued)

Statistical Associations between Enrollment and Actual NSS (per foundation pupil)

Key Finding

District size and enrollment cohort show associations with local funding effort patterns. This section examines Actual Net School Spending above Required NSS (per foundation pupil) - a measure of spending beyond state-mandated minimums.

Distribution of Actual NSS above Required NSS by Enrollment Cohort

The five-number summaries below show: current local funding effort (2024 values) and how this effort changed over time (2009-2024 growth rates).

2024 Actual NSS above Required NSS per Foundation Pupil by Cohort

Box plots show the spread of spending above requirements within each size cohort. Positive values indicate spending above the state-required minimum; negative values indicate spending below required levels. The median line shows typical local effort for each cohort.

Cohort	n	Min	Q1	Median	Q3	Max	Distribution
Tiny (0-200 FTE)	15	$1,085	$7,274	$10,086	$12,440	$23,329	
Small (201-800 FTE)	10	$165	$1,980	$6,233	$8,230	$11,848	
Medium (801-1600 FTE)	8	$629	$1,475	$3,563	$6,467	$11,670	
Large (1601-10K FTE)	11	$-526	$1,198	$2,875	$4,983	$5,171	
Outliers (Springfield >10K FTE)	1	$-0	$-0	$-0	$-0	$-0	

2009-2024 Actual NSS above Required NSS Growth Rates by Cohort

These box plots reveal whether districts of similar size increased local effort at similar rates (narrow boxes) or diverged over time (wide boxes). Comparing medians shows which cohorts increased spending above requirements faster or slower.

Cohort	n	Min	Q1	Median	Q3	Max	Distribution
Tiny (0-200 FTE)	15	-8.7%	3.6%	5.1%	7.3%	18.3%	
Small (201-800 FTE)	10	-9.9%	2.4%	5.4%	9.0%	19.4%	
Medium (801-1600 FTE)	8	-0.6%	2.6%	3.6%	5.6%	16.3%	
Large (1601-10K FTE)	11	-0.8%	4.3%	7.4%	10.5%	14.1%	
Outliers (Springfield >10K FTE)	1	59.5%	59.5%	59.5%	59.5%	59.5%	

Section 1 — Western MA traditional public school district trends (continued)

Statistical Test Results for Actual NSS above Required NSS (per foundation pupil)

Statistical Test Results for Actual NSS above Required NSS

What Associates with Current Local Funding Effort (2024 Actual NSS above Required NSS)?

District size (cohort) — MODERATE ASSOCIATION

• ANOVA: $F(3,36) = 4.73$, $p = 0.007$, $\eta^2 = 0.283$
• Enrollment cohort explains ~28% of local funding effort variation
• Effect size: medium (wealth and tax base often correlate with district size)

Enrollment growth rate — NO ASSOCIATION

• Linear regression: $r = 0.087$, $R^2 = 0.008$, $p = 0.607$
• Explains less than 1% of funding effort variation
• Not statistically significant at $\alpha = 0.05$

Combined model — MODEST PREDICTIVE POWER

• ANCOVA: $R^2 = 0.291$, Adjusted $R^2 = 0.210$, $F(4,35) = 4.12$, $p = 0.008$
• Size + growth together explain 29% of variation
• Enrollment growth adds no predictive value once cohort is accounted for

What Associates with Local Funding Effort Growth (2009-2024 CAGR)?

District size (cohort) — NO ASSOCIATION

• ANOVA: $F(3,36) = 0.52$, $p = 0.669$, $\eta^2 = 0.042$
• Effect size: negligible
• All cohorts show similar growth rate distributions in local funding effort

Enrollment growth rate — NO ASSOCIATION

• Linear regression: $r = -0.068$, $R^2 = 0.005$, $p = 0.689$
• Weak negative correlation (not significant)
• Declining enrollment doesn't predict changes in funding effort

Combined model — NO PREDICTIVE POWER

• ANCOVA: $R^2 = 0.047$, Adjusted $R^2 = -0.062$, $F(4,35) = 0.51$, $p = 0.730$
• Negative adjusted R^2 indicates model performs worse than sample mean
• Enrollment characteristics don't predict local funding trajectories

Interpretation

These analyses examine statistical associations, not causal relationships. The weak associations demonstrate that:

• Local funding effort varies widely among districts of similar size
• Enrollment change doesn't systematically drive changes in spending above requirements
• Community wealth, property values, governance decisions, and historical commitments matter more than enrollment characteristics

Section 2 — Western MA cohort details

Section 2 provides detailed per-pupil expenditure (PPE) and Net School Spending (NSS) analysis for Western Massachusetts enrollment cohorts: Tiny, Small, Medium, Large, and Springfield. Each cohort section includes:

- Aggregate **expenditure trends by category** (instruction, administration, operations, etc.)
- Aggregate **Chapter 70 aid** and **NSS** patterns over time
- **Enrollment-weighted averages** that represent typical spending across all districts in each cohort

These cohort pages serve as regional baselines for understanding cost patterns by district size.

To skip to **Section 3 — Selected districts**, turn to page 35.

Report Navigation:

Section 2 — Western MA cohort details

Tiny (0-200 FTE) — PPE and Enrollment — Weighted average per district.

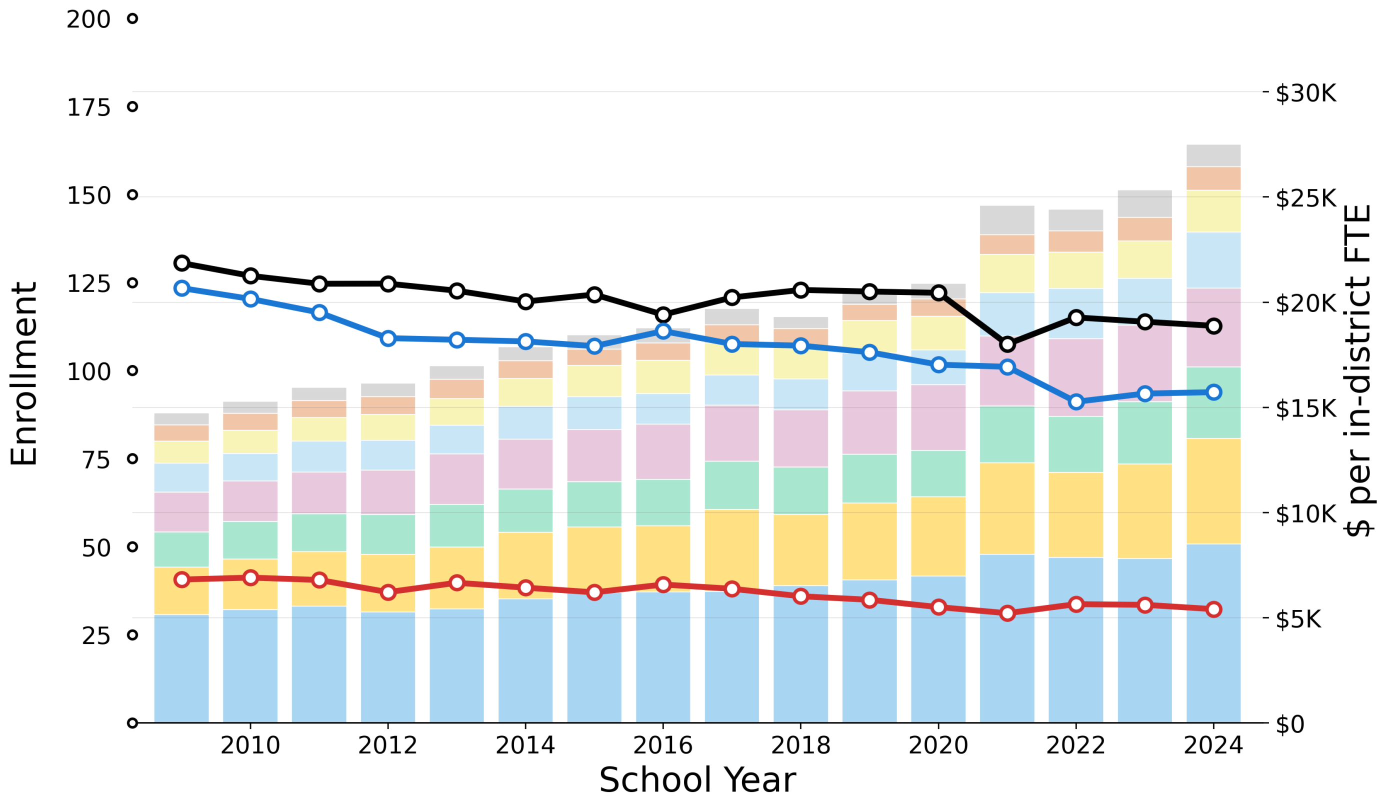


Figure 11

Table 11

Category	2009 $/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 $/pupil
Other	$577	+4.1%	+4.7%	+9.4%	$1,057
Administration	$770	+2.6%	+2.8%	+7.9%	$1,126
Instructional Leadership	$1,042	+4.4%	+4.2%	+5.2%	$1,991
Operations and Maintenance	$1,374	+4.5%	+5.5%	+7.9%	$2,652
Other Teaching Services	$1,899	+4.7%	+4.7%	+4.6%	$3,757
Pupil Services	$1,669	+4.9%	+5.2%	+8.0%	$3,406
Insurance, Retirement and Other	$2,258	+5.5%	+4.8%	+6.5%	$5,009
Teachers	$5,148	+3.4%	+3.7%	+4.6%	$8,507
Total	$14,738	+4.2%	+4.4%	+6.0%	$27,504

Table 12

In-district FTE	2009	CAGR 15y	CAGR 10y	CAGR 5y	2024
In-District FTE Pupils	131	-1.0%	-0.6%	-1.6%	113
Foundation Enrollment	123	-1.8%	-1.4%	-2.3%	94
Out-of-District FTE Pupils	41	-1.5%	-1.7%	-1.6%	32

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔEnrollment| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp	Above Western MA baseline
CAGR = (End/Start)^(1/years) – 1	Below Western MA baseline

→ *Compare to PPE and Enrollment for Western MA (all, excl. Springfield) (page 33), Leverett (page 40), Pelham (page 42), Shutesbury (page 44)*

Section 2 — Western MA cohort details

Tiny (0-200 FTE) — Chapter 70 Aid and Net School Spending (per foundation pupil).

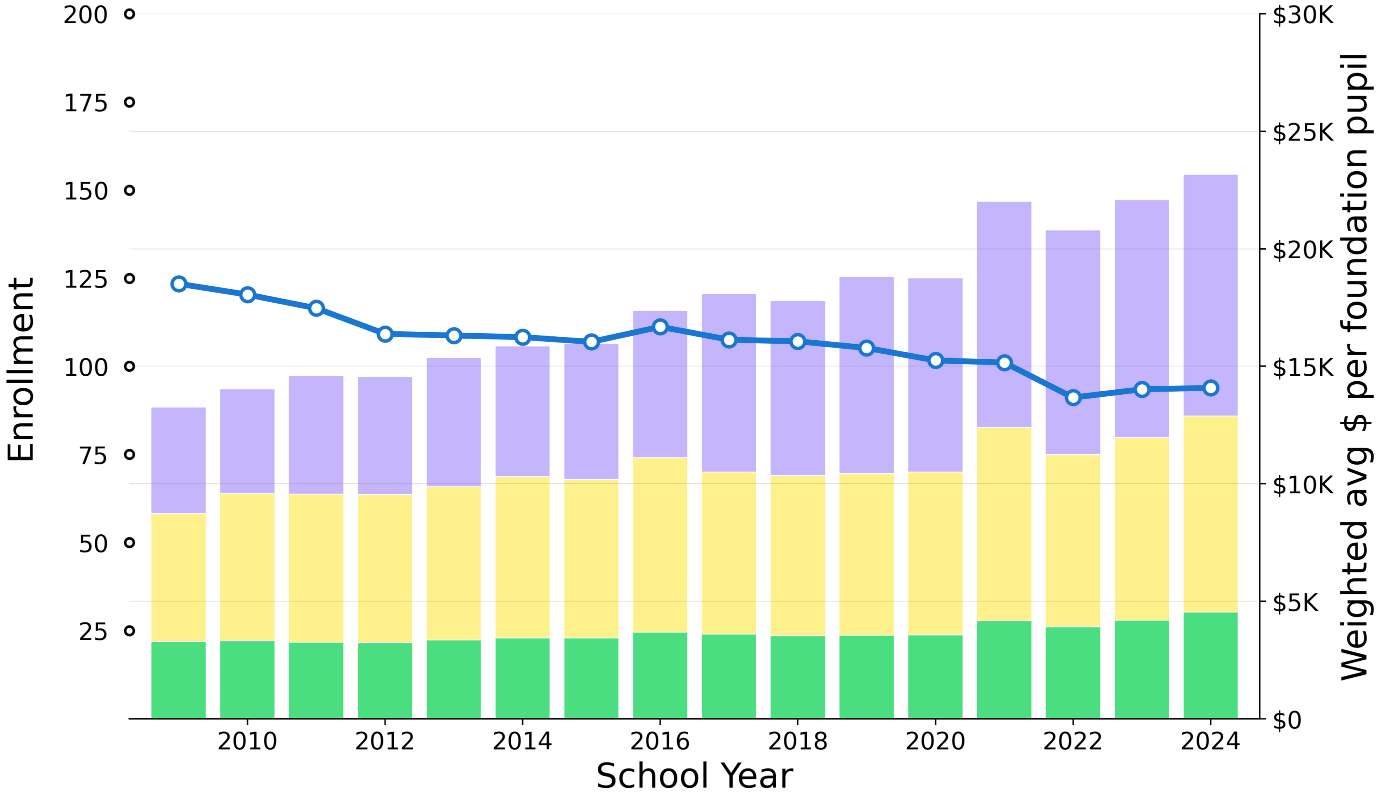


Figure 12

Table 13

	Component	2009 $/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 $/pupil
	Actual NSS (minus Req NSS)	$4,523	5.63%	6.32%	4.13%	$10,285
	Req NSS (minus Ch70)	$5,438	2.89%	1.98%	3.95%	$8,338
	Ch70 Aid	$3,297	2.17%	2.82%	5.02%	$4,547
	Total (Actual NSS)	$13,258	3.79%	3.86%	4.23%	$23,170

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp — Above Western MA baseline

CAGR = (End/Start)^(1/years) – 1 — Below Western MA baseline

Tiny (0-200 FTE) includes 17 members:

Districts (10): Erving, Florida, Hancock, Leverett, Pelham, Richmond, Rowe, Savoy, Shutesbury, Worthington.

Regional K-12 (7): Conway, Farmington River Reg, Hawlemont, New Salem-Wendell, Petersham, Sunderland, Whately.

→ *Compare to Chapter 70 Aid and Net School Spending for Western MA (all, excl. Springfield) (page 34), Leverett (page 41), Pelham (page 43), Shutesbury (page 45)*

Section 2 — Western MA cohort details

Small (201-800 FTE) — PPE and Enrollment — Weighted average per district.

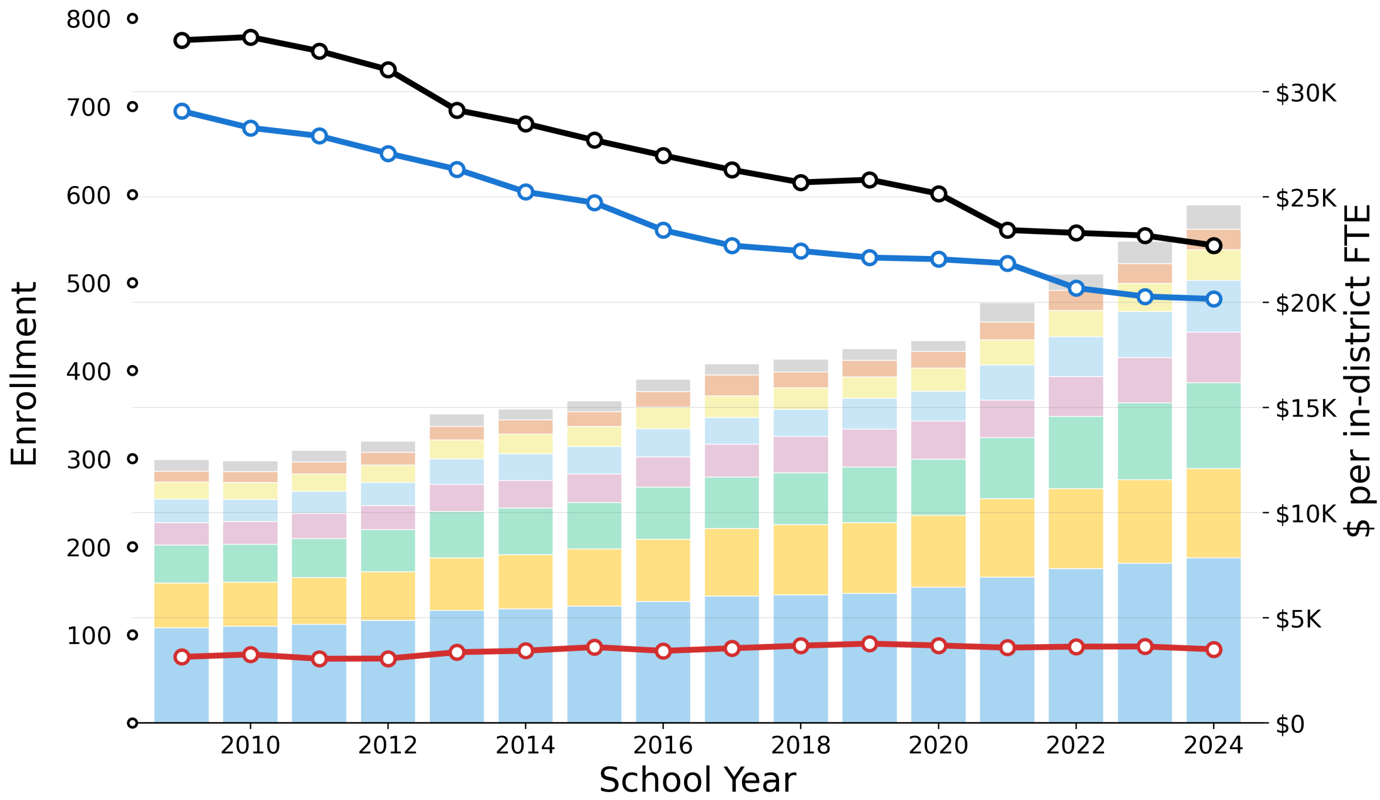


Figure 13

Table 14

Category	2009 $/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 $/pupil
Other	$558	+4.9%	+8.4%	+15.6%	$1,150
Administration	$523	+4.3%	+3.9%	+4.4%	$976
Instructional Leadership	$811	+4.0%	+4.4%	+7.6%	$1,463
Operations and Maintenance	$1,108	+5.4%	+6.9%	+10.8%	$2,454
Other Teaching Services	$1,061	+5.6%	+6.2%	+6.0%	$2,402
Pupil Services	$1,813	+5.6%	+6.3%	+9.1%	$4,084
Insurance, Retirement and Other	$2,128	+4.7%	+5.2%	+4.8%	$4,254
Teachers	$4,522	+3.7%	+3.8%	+5.0%	$7,838
Total	$12,523	+4.6%	+5.1%	+6.7%	$24,622

Table 15

In-district FTE	2009	CAGR 15y	CAGR 10y	CAGR 5y	2024
In-District FTE Pupils	775	-2.4%	-2.2%	-2.5%	542
Foundation Enrollment	695	-2.4%	-2.2%	-1.8%	481
Out-of-District FTE Pupils	75	+0.7%	+0.2%	-1.5%	84

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔEnrollment| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp	Above Western MA baseline
CAGR = (End/Start)^(1/years) – 1	Below Western MA baseline

Section 2 — Western MA cohort details

Small (201-800 FTE) — Chapter 70 Aid and Net School Spending (per foundation pupil).

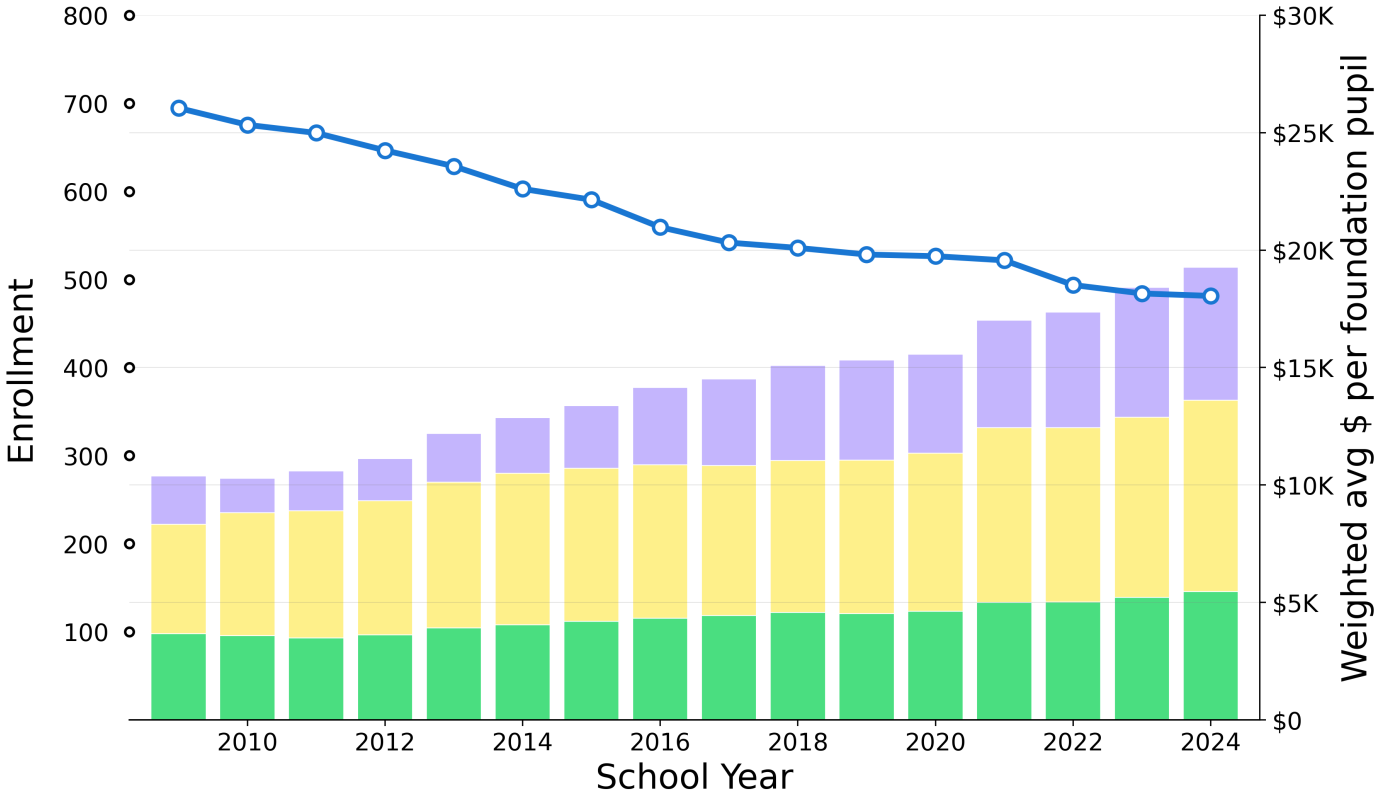


Figure 14

Table 16

	Component	2009 $/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 $/pupil
	Actual NSS (minus Req NSS)	$2,054	6.98%	9.11%	5.86%	$5,656
	Req NSS (minus Ch70)	$4,657	3.79%	2.35%	4.52%	$8,139
	Ch70 Aid	$3,672	2.69%	3.05%	3.81%	$5,465
	Total (Actual NSS)	$10,384	4.20%	4.12%	4.69%	$19,261

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp — Above Western MA baseline

CAGR = (End/Start)^(1/years) – 1 — Below Western MA baseline

Small (201-800 FTE) includes 13 members:

Districts (6): Clarksburg, Granby, Hadley, Hatfield, Lee, Lenox.

Regional K-12 (5): Deerfield, Gateway, Orange, Pioneer Valley, Southern Berkshire.

Secondary regions (2): Frontier, Ralph C Mahar.

Section 2 — Western MA cohort details

Medium (801-1600 FTE) — PPE and Enrollment — Weighted average per district.

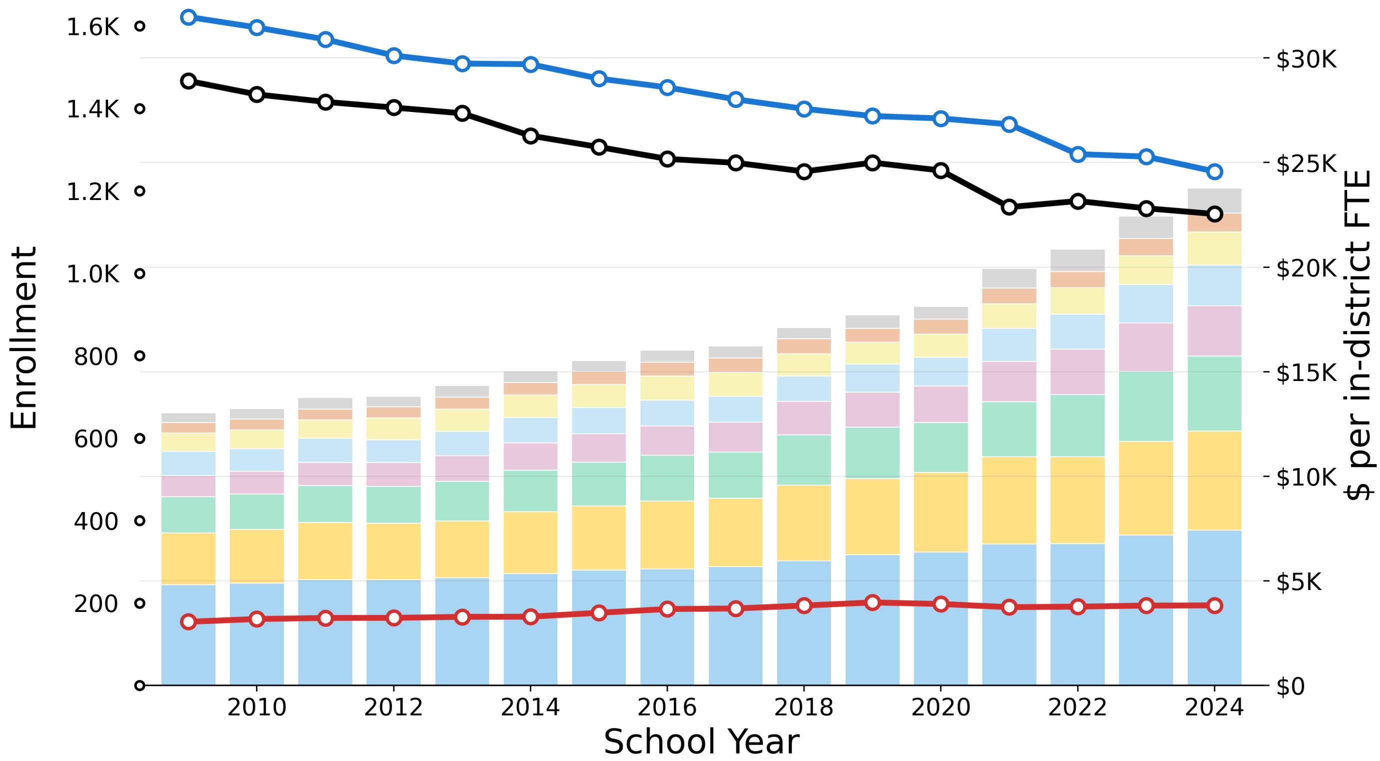


Figure 15 *Table 17*

	Category	2009 $/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 $/pupil
	Other	$461	+6.6%	+8.1%	+13.3%	$1,200
	Administration	$494	+4.2%	+4.7%	+6.9%	$911
	Instructional Leadership	$893	+3.8%	+3.7%	+8.2%	$1,573
	Operations and Maintenance	$1,150	+3.6%	+4.9%	+8.0%	$1,953
	Other Teaching Services	$1,014	+5.9%	+6.2%	+7.3%	$2,395
	Pupil Services	$1,724	+5.0%	+6.1%	+7.8%	$3,581
	Insurance, Retirement and Other	$2,483	+4.4%	+4.9%	+5.5%	$4,740
	Teachers	$4,813	+2.9%	+3.3%	+3.5%	$7,431
	Total	$13,032	+4.1%	+4.7%	+6.1%	$23,785

Table 18

In-district FTE	2009	CAGR 15y	CAGR 10y	CAGR 5y	2024
In-District FTE Pupils	1,466	-1.6%	-1.5%	-2.0%	1,144
Foundation Enrollment	1,622	-1.7%	-1.9%	-2.0%	1,247
Out-of-District FTE Pupils	154	+1.5%	+1.5%	-0.7%	194

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔEnrollment| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp	Above Western MA baseline
CAGR = (End/Start)^(1/years) – 1	Below Western MA baseline

→ *Compare to PPE and Enrollment for Western MA (all, excl. Springfield) (page 33), Amherst-Pelham (page 36), Amherst (page 38)*

Section 2 — Western MA cohort details

Medium (801-1600 FTE) — Chapter 70 Aid and Net School Spending (per foundation pupil).

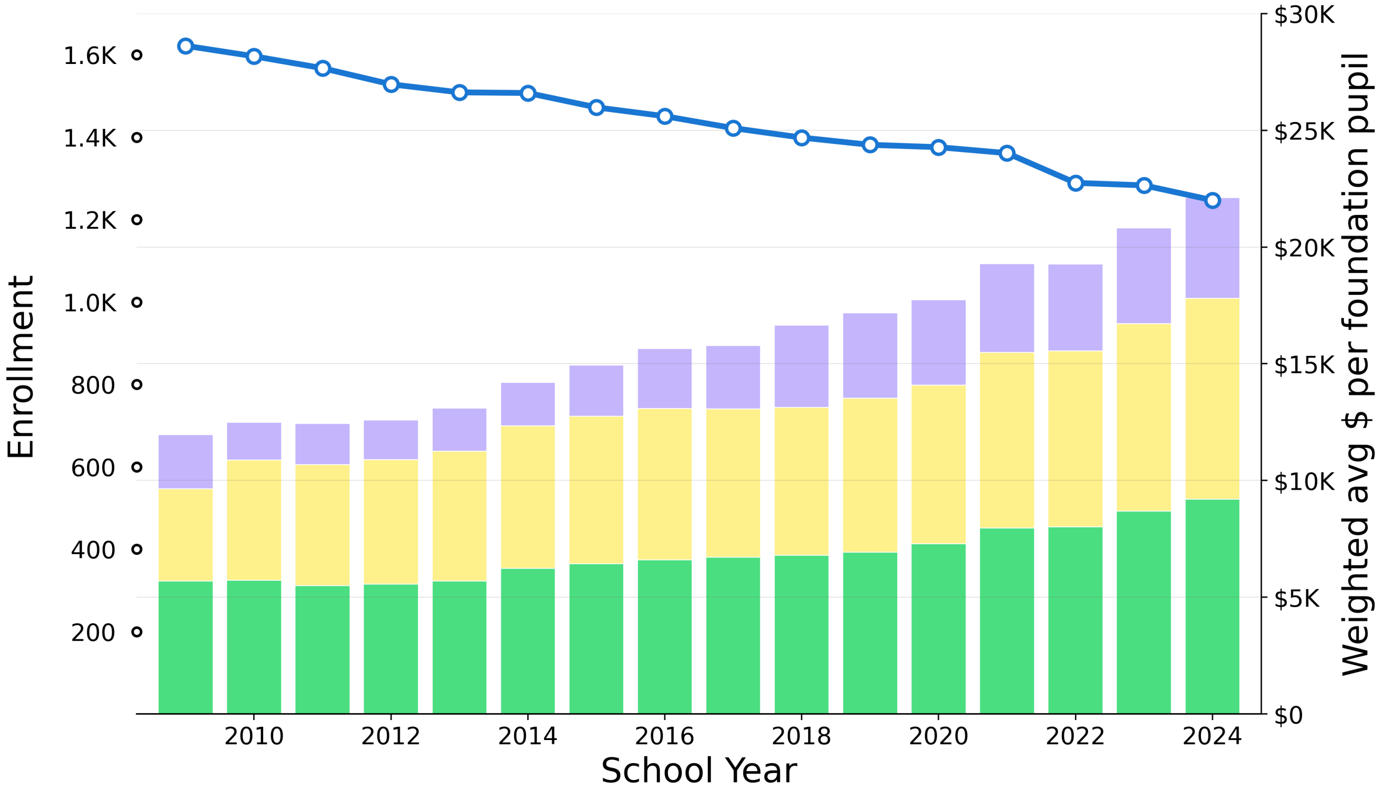


Figure 16

Table 19

	Component	2009 $/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 $/pupil
	Actual NSS (minus Req NSS)	$2,318	4.23%	8.76%	3.35%	$4,315
	Req NSS (minus Ch70)	$3,945	5.34%	3.49%	5.50%	$8,605
	Ch70 Aid	$5,698	3.25%	3.97%	5.82%	$9,202
	Total (Actual NSS)	$11,961	4.18%	4.53%	5.19%	$22,121

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp — Above Western MA baseline

CAGR = (End/Start)^(1/years) – 1 — Below Western MA baseline

Medium (801-1600 FTE) includes 15 members:

Districts (8): Amherst, Berkshire Hills, Easthampton, Greenfield, Monson, North Adams, Palmer, Ware.

Regional K-12 (6): Central Berkshire, Gill-Montague, Hoosac Valley Regional, Mohawk Trail, Mount Greylock, Southwick-Tolland-Granville Regional School District.

Secondary regions (1): Amherst-Pelham.

→ Compare to Chapter 70 Aid and Net School Spending for Western MA (all, excl. Springfield) (page 34), Amherst-Pelham (page 37), Amherst (page 39)

Section 2 — Western MA cohort details

Large (1601-10K FTE) — PPE and Enrollment — Weighted average per district.

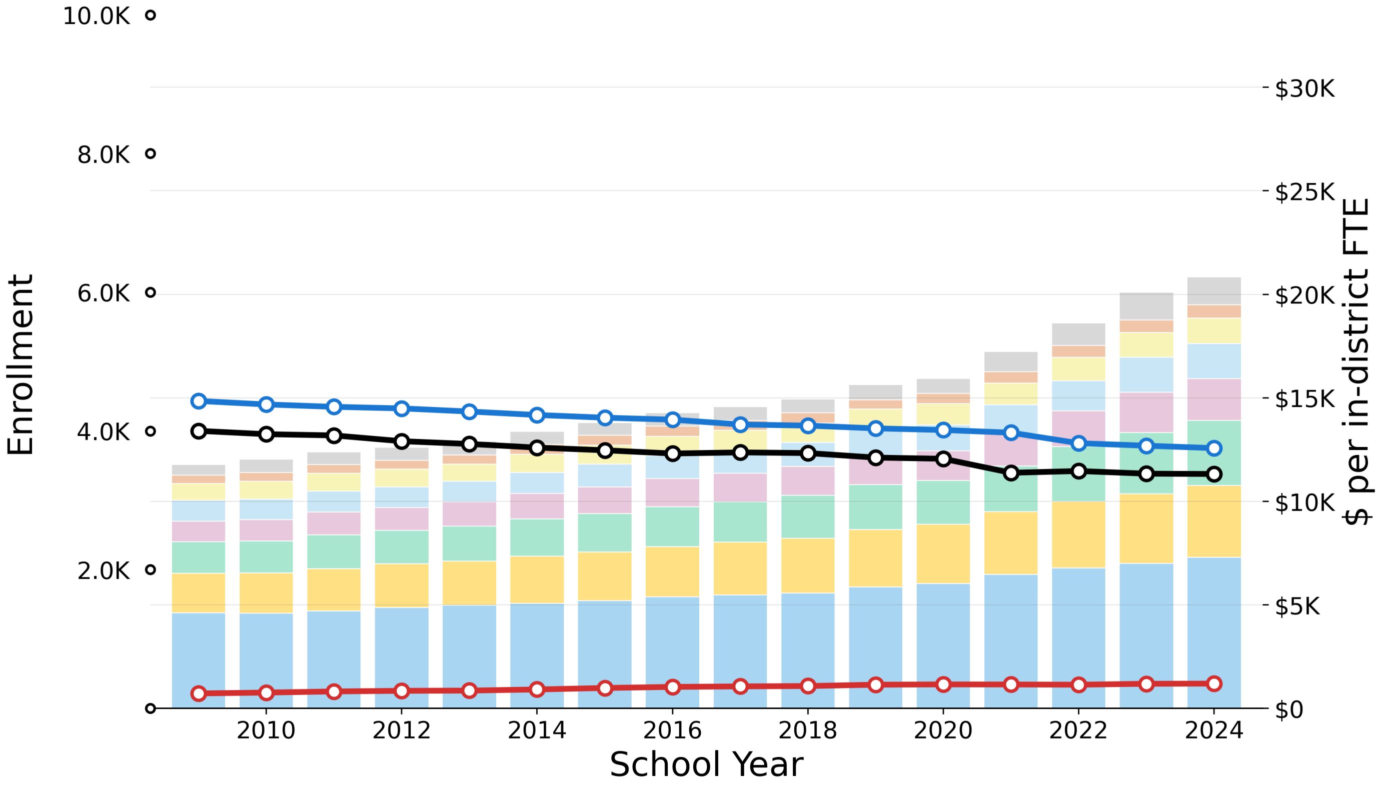


Figure 17

Table 20

	Category	2009 $/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 $/pupil
	Other	$513	+6.6%	+7.9%	+13.3%	$1,348
	Administration	$392	+3.4%	+3.5%	+7.5%	$649
	Instructional Leadership	$811	+2.8%	+3.3%	+4.3%	$1,231
	Operations and Maintenance	$1,014	+3.4%	+5.1%	+6.2%	$1,683
	Other Teaching Services	$990	+4.9%	+5.1%	+7.5%	$2,021
	Pupil Services	$1,531	+4.9%	+5.8%	+7.7%	$3,148
	Insurance, Retirement and Other	$1,911	+4.1%	+4.3%	+4.6%	$3,469
	Teachers	$4,613	+3.1%	+3.7%	+4.5%	$7,292
	Total	$11,775	+3.9%	+4.5%	+5.9%	$20,841

Table 21

	In-district FTE	2009	CAGR 15y	CAGR 10y	CAGR 5y	2024
	In-District FTE Pupils	4,001	-1.1%	-1.1%	-1.3%	3,379
	Foundation Enrollment	4,432	-1.1%	-1.2%	-1.5%	3,752
	Out-of-District FTE Pupils	215	+3.4%	+2.7%	+0.9%	357

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔEnrollment| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp	Above Western MA baseline
CAGR = (End/Start)^(1/years) – 1	Below Western MA baseline

Section 2 — Western MA cohort details

Large (1601-10K FTE) — Chapter 70 Aid and Net School Spending (per foundation pupil).

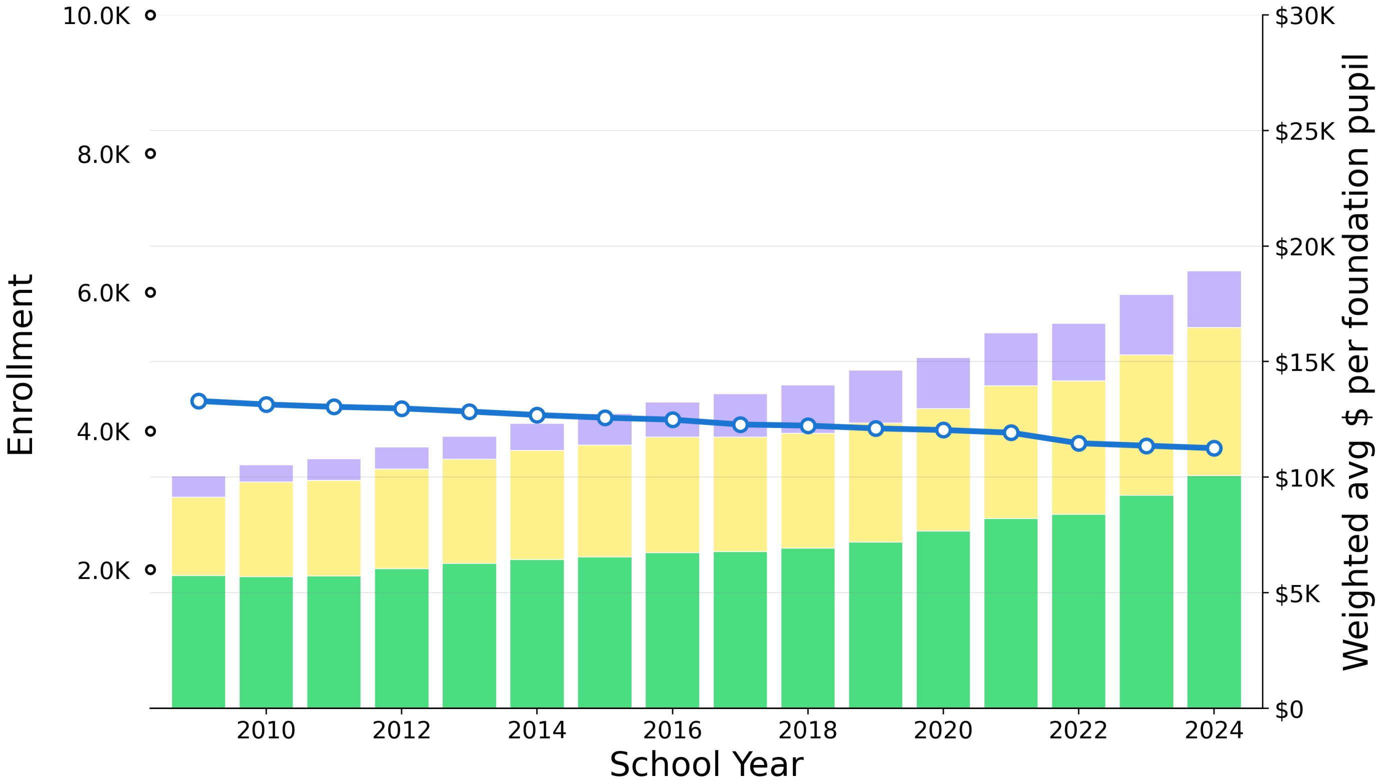


Figure 18

Table 22

	Component	2009 $/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 $/pupil
	Actual NSS (minus Req NSS)	$934	6.66%	7.71%	1.41%	$2,455
	Req NSS (minus Ch70)	$3,385	4.35%	3.11%	4.46%	$6,407
	Ch70 Aid	$5,742	3.81%	4.57%	6.96%	$10,065
	Total (Actual NSS)	$10,060	4.30%	4.39%	5.29%	$18,927

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp — Above Western MA baseline

CAGR = (End/Start)^(1/years) – 1 — Below Western MA baseline

Large (1601-10K FTE) includes 14 members:

Districts (12): Agawam, Belchertown, Chicopee, East Longmeadow, Holyoke, Longmeadow, Ludlow, Northampton, Pittsfield, South Hadley, West Springfield, Westfield.

Regional K-12 (2): Athol-Royalston, Hampden-Wilbraham.

Section 2 — Western MA cohort details

Outliers (Springfield >10K FTE) — PPE and Enrollment — Weighted average per district.

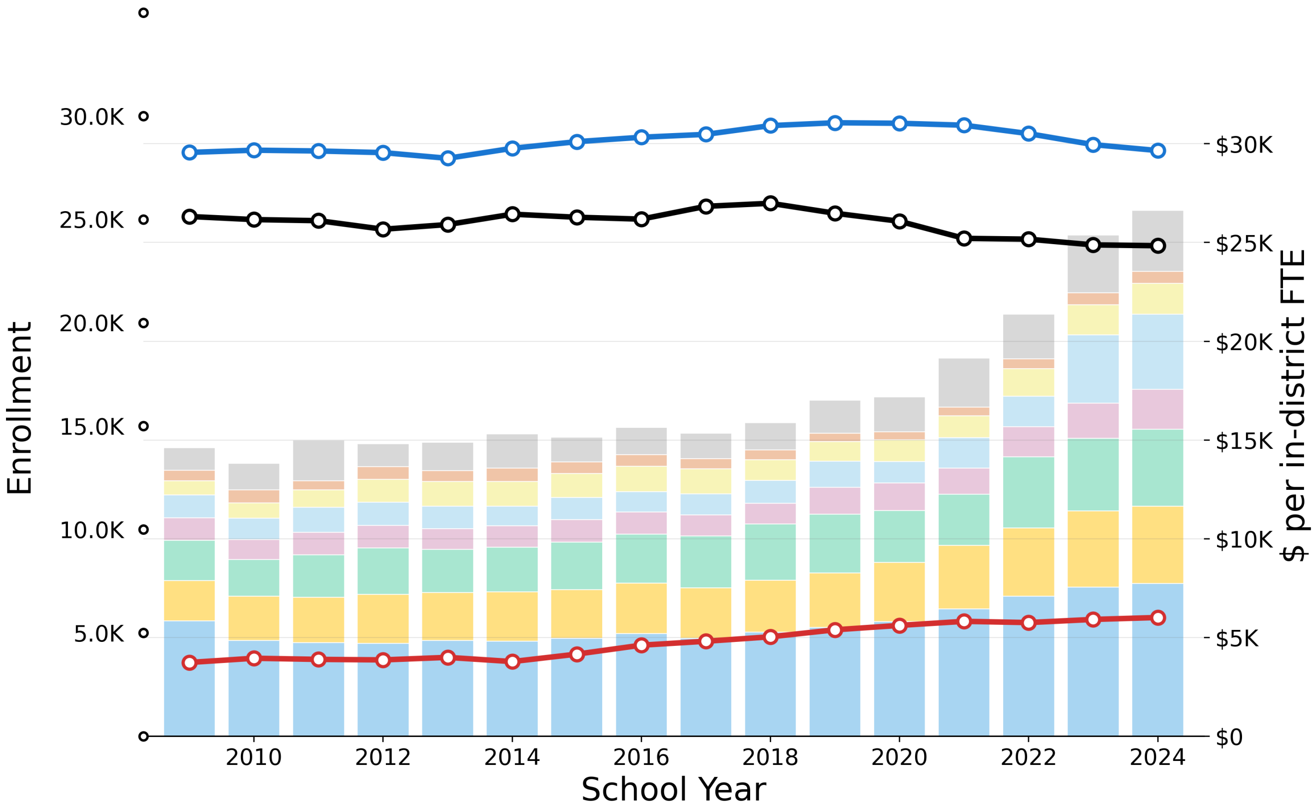


Figure 19

Table 23

Category	2009 $/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 $/pupil
Other	$1,143	+6.8%	+5.9%	+12.9%	$3,071
Administration	$533	+0.9%	-1.1%	+7.9%	$609
Instructional Leadership	$719	+5.3%	+2.4%	+9.8%	$1,571
Operations and Maintenance	$1,150	+8.3%	+14.3%	+23.3%	$3,806
Other Teaching Services	$1,143	+3.8%	+6.4%	+8.2%	$2,014
Pupil Services	$2,031	+4.4%	+5.6%	+5.6%	$3,901
Insurance, Retirement and Other	$2,050	+4.4%	+4.6%	+7.2%	$3,906
Teachers	$5,839	+1.9%	+4.8%	+7.0%	$7,737
Total	$14,608	+4.1%	+5.7%	+9.4%	$26,615

Table 24

In-district FTE	2009	CAGR 15y	CAGR 10y	CAGR 5y	2024
In-District FTE Pupils	25,137	-0.4%	-0.6%	-1.3%	23,723
Foundation Enrollment	28,235	+0.0%	-0.0%	-0.9%	28,326
Out-of-District FTE Pupils	3,571	+3.2%	+4.7%	+2.2%	5,745

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔEnrollment| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp	Above Western MA baseline
CAGR = (End/Start)^(1/years) − 1	Below Western MA baseline

Section 2 — Western MA cohort details

Outliers (Springfield >10K FTE) — Chapter 70 Aid and Net School Spending (per foundation pupil).

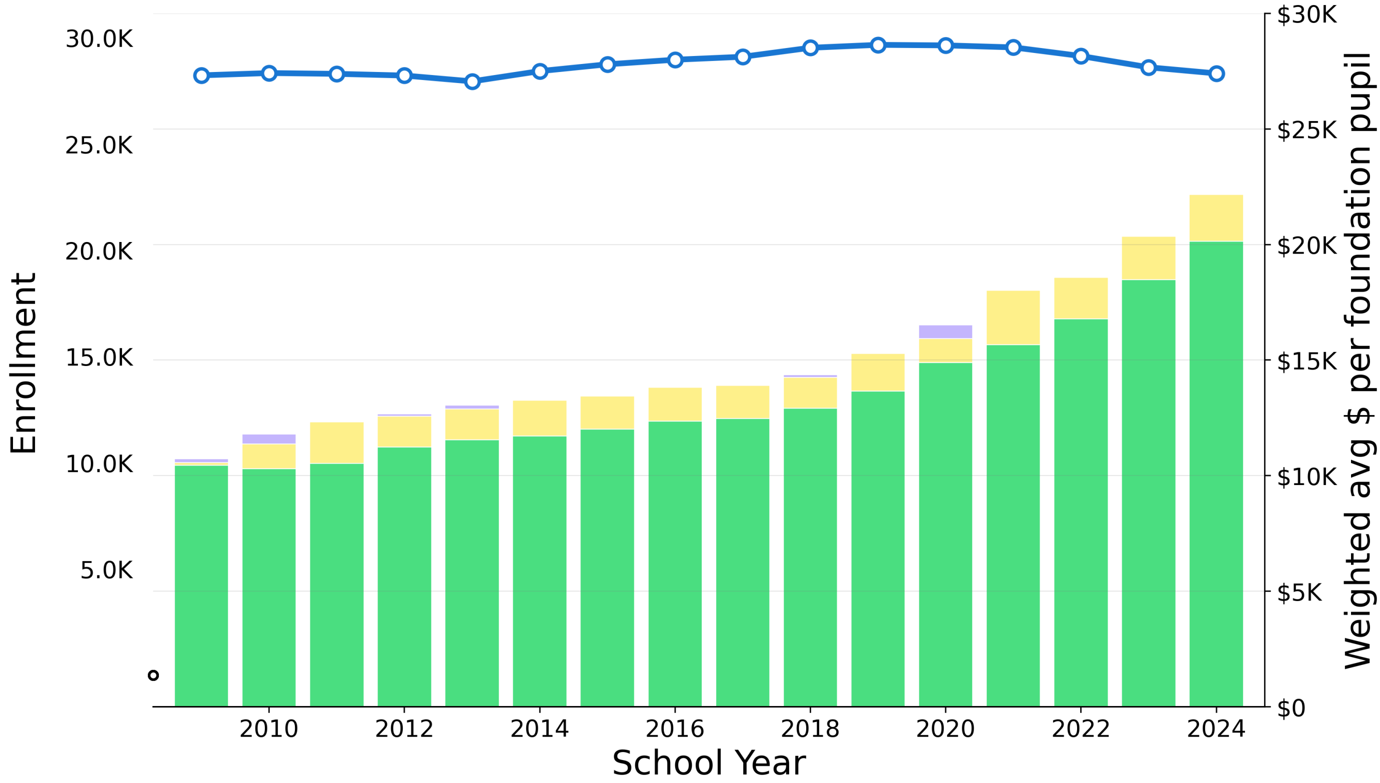


Figure 20

Table 25

Component	2009 $/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 $/pupil
Actual NSS (minus Req NSS)	$-169	-59.51%	-52.40%	24.01%	$-0
Req NSS (minus Ch70)	$278	14.15%	2.69%	4.35%	$2,024
Ch70 Aid	$10,452	4.47%	5.57%	8.08%	$20,138
Total (Actual NSS)	$10,561	5.07%	5.26%	7.70%	$22,163

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp — Above Western MA baseline

CAGR = (End/Start)^(1/years) – 1 — Below Western MA baseline

Outliers (Springfield >10K FTE) includes 1 members:

Districts (1): Springfield.

Section 2 — Western MA cohort details

Western MA (all, excl. Springfield) — PPE and Enrollment

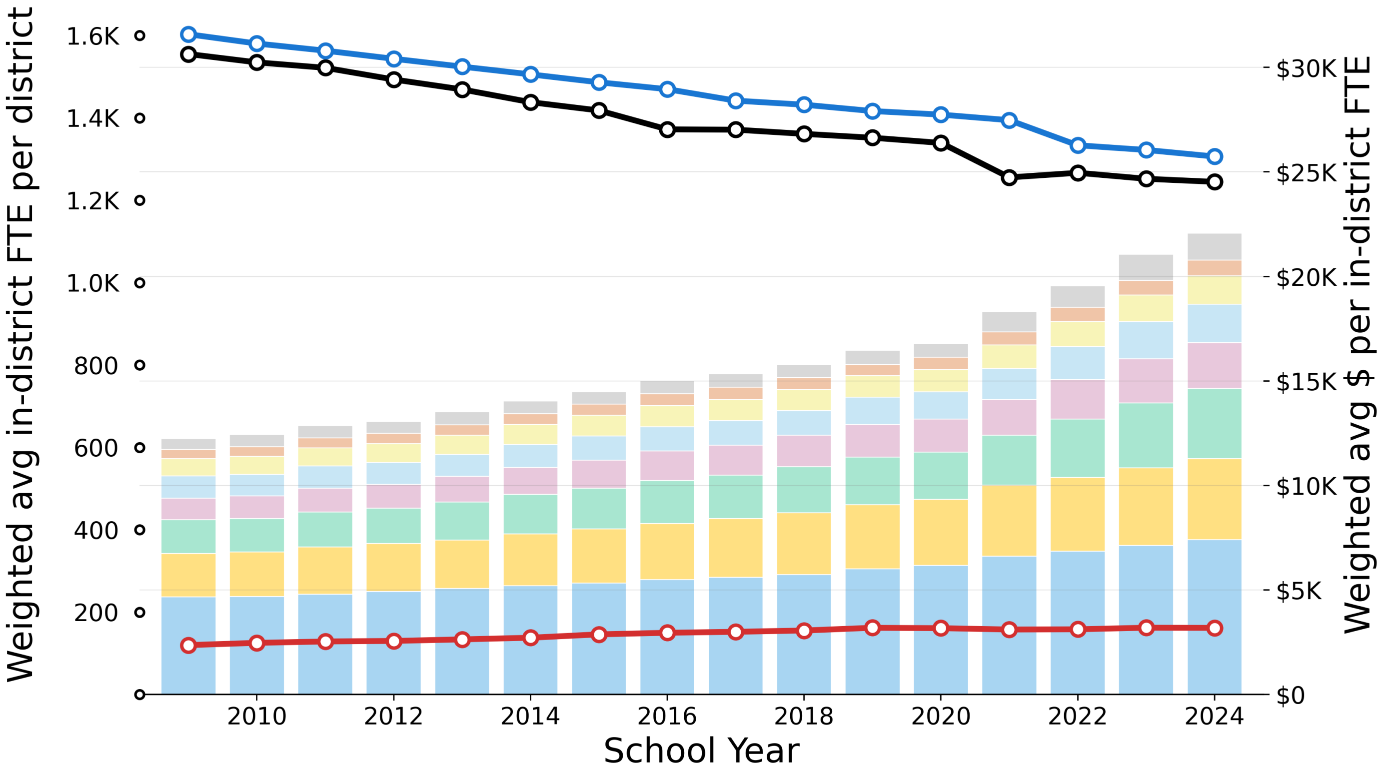


Figure 21

Table 26

Category	2009 $/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 $/pupil
Other	$507	+6.4%	+7.9%	+13.4%	$1,287
Administration	$440	+3.6%	+3.8%	+6.8%	$754
Instructional Leadership	$836	+3.3%	+3.6%	+5.6%	$1,353
Operations and Maintenance	$1,066	+3.7%	+5.3%	+7.2%	$1,845
Other Teaching Services	$1,025	+5.2%	+5.5%	+7.1%	$2,190
Pupil Services	$1,613	+5.0%	+5.9%	+7.9%	$3,346
Insurance, Retirement and Other	$2,083	+4.2%	+4.5%	+4.9%	$3,881
Teachers	$4,664	+3.1%	+3.6%	+4.3%	$7,409
Total	$12,234	+4.0%	+4.6%	+6.0%	$22,065

Table 27

In-district FTE	2009	CAGR 15y	CAGR 10y	CAGR 5y	2024
In-District FTE Pupils	1,555	-1.5%	-1.4%	-1.6%	1,245
Foundation Enrollment	1,603	-1.4%	-1.4%	-1.6%	1,306
Out-of-District FTE Pupils	120	+2.0%	+1.6%	-0.0%	162

→ Compare to PPE and Enrollment for Tiny (0-200 FTE) cohort (page 23), Small (201-800 FTE) cohort (page 25), Medium (801-1600 FTE) cohort (page 27), Large (1601-10K FTE) cohort (page 29), Outliers (Springfield >10K FTE) cohort (page 31)

Section 2 — Western MA cohort details

Western MA (all, excl. Springfield) — Chapter 70 Aid and Net School Spending (per foundation pupil)

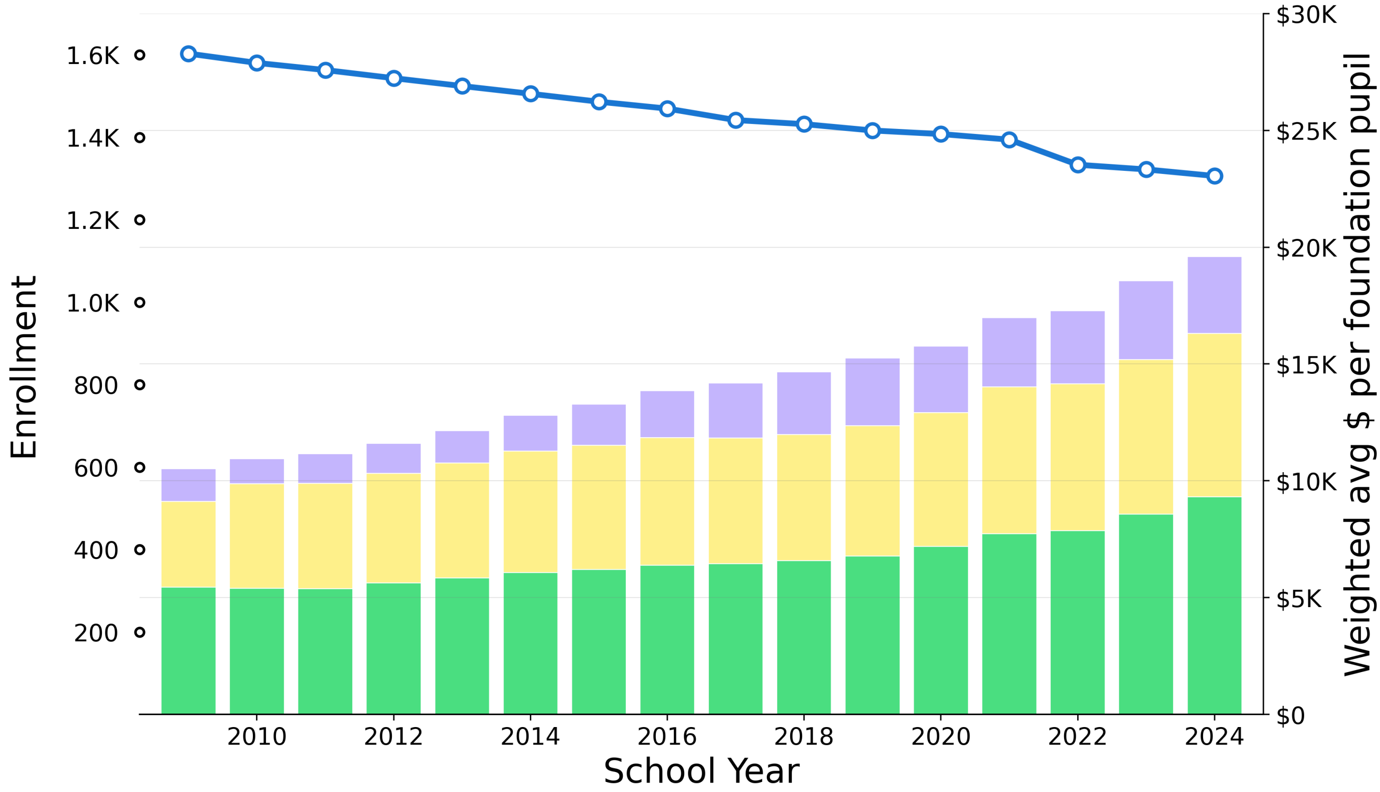


Figure 22

Table 28

	Component	2009 $/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 $/pupil
	Actual NSS (minus Req NSS)	$1,394	5.91%	8.00%	2.64%	$3,301
	Req NSS (minus Ch70)	$3,676	4.38%	3.02%	4.60%	$6,988
	Ch70 Aid	$5,444	3.64%	4.36%	6.57%	$9,313
	Total (Actual NSS)	$10,515	4.24%	4.36%	5.14%	$19,601

Western MA (all, excl. Springfield) includes 59 members:

Districts (36): Agawam, Amherst, Belchertown, Berkshire Hills, Chicopee, Clarksburg, East Longmeadow, Easthampton, Erving, Florida, Granby, Greenfield, Hadley, Hancock, Hatfield, Holyoke, Lee, Lenox, Leverett, Longmeadow, Ludlow, Monson, North Adams, Northampton, Palmer, Pelham, Pittsfield, Richmond, Rowe, Savoy, Shutesbury, South Hadley, Ware, West Springfield, Westfield, Worthington.

Regional K-12 (20): Athol-Royalston, Central Berkshire, Conway, Deerfield, Farmington River Reg, Gateway, Gill-Montague, Hampden-Wilbraham, Hawlemont, Hoosac Valley Regional, Mohawk Trail, Mount Greylock, New Salem-Wendell, Orange, Petersham, Pioneer Valley, Southern Berkshire, Southwick-Tolland-Granville Regional School District, Sunderland, Whately.

Secondary regions (3): Amherst-Pelham, Frontier, Ralph C Mahar.

Note: The following districts are omitted from this analysis: Chesterfield-Goshen (missing expenditure data), Hampshire (missing expenditure data), Southampton (missing expenditure data), Warwick (no enrollment data), Westhampton (missing expenditure data), Williamsburg (missing expenditure data).

→ Compare to Chapter 70 Aid and Net School Spending for Tiny (0-200 FTE) cohort (page 24), Small (201-800 FTE) cohort (page 26), Medium (801-1600 FTE) cohort (page 28), Large (1601-10K FTE) cohort (page 30), Outliers (Springfield >10K FTE) cohort (page 32)

Section 3 — Selected districts

Section 3 examines five selected districts in depth: Amherst-Pelham Regional, Amherst, Leverett, Pelham, and Shutesbury. For each district, three detailed pages show:

- Total PPE breakdown by **expenditure category** with shaded comparison to cohort baseline
- **Chapter 70 aid** and **Net School Spending** breakdown per pupil with shaded comparison to cohort baseline
- **CAGR over 5, 10, and 15-year periods** with shaded comparison to cohort baseline

These district-level pages enable direct comparison between individual district spending patterns and their respective cohorts. Note that although this section focuses on the five selected districts and Western MA, the same visualizations can be produced for other MA districts and regions.

For detailed methodology, see **Appendix A: Data Sources & Calculation Methodology** (https://github.com/timshores/schools/blob/main/ppe_report/output/WMPPE%20Appendices.pdf).

Report Navigation:

Section 3 — Specific districts

Amherst-Pelham Regional — PPE and Enrollment

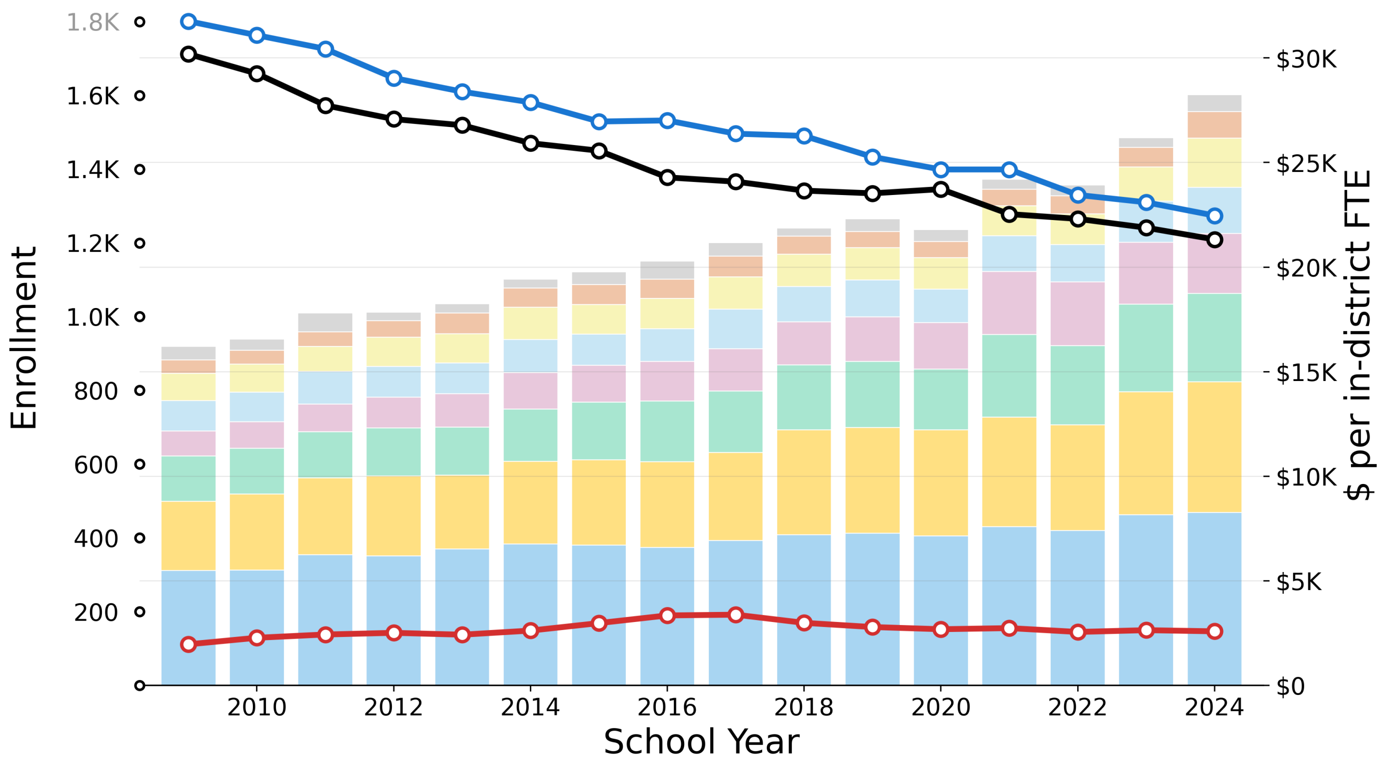


Figure 23

Table 29

	Category	2009 $/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 $/pupil
	Other	$646	+1.4%	+6.5%	+5.5%	$799
	Administration	$652	+4.6%	+3.5%	+11.0%	$1,277
	Instructional Leadership	$1,289	+4.0%	+4.2%	+8.6%	$2,336
	Operations and Maintenance	$1,459	+2.8%	+3.4%	+4.6%	$2,221
	Other Teaching Services	$1,199	+5.9%	+5.1%	+6.1%	$2,851
	Pupil Services	$2,170	+4.6%	+5.4%	+6.0%	$4,231
	Insurance, Retirement and Other	$3,305	+4.3%	+4.7%	+4.4%	$6,248
	Teachers	$5,491	+2.8%	+2.0%	+2.6%	$8,270
	Total	$16,211	+3.8%	+3.8%	+4.8%	$28,233

Table 30

In-district FTE	2009	CAGR 15y	CAGR 10y	CAGR 5y	2024
In-District FTE Pupils	1,712	-2.3%	-1.9%	-1.9%	1,209
Foundation Enrollment	1,801	-2.3%	-2.1%	-2.3%	1,274
Out-of-District FTE Pupils	112	+1.8%	-0.1%	-1.5%	147

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔEnrollment| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp	Above cohort baseline (Medium (801-1600))
CAGR = (End/Start)^(1/years) – 1	Below cohort baseline (Medium (801-1600))

→ *Compare to Tiny (0-200 FTE) cohort: PPE and Enrollment (page 23)*

Section 3 — Selected districts

Amherst-Pelham Regional — Chapter 70 Aid and Net School Spending (per foundation pupil)

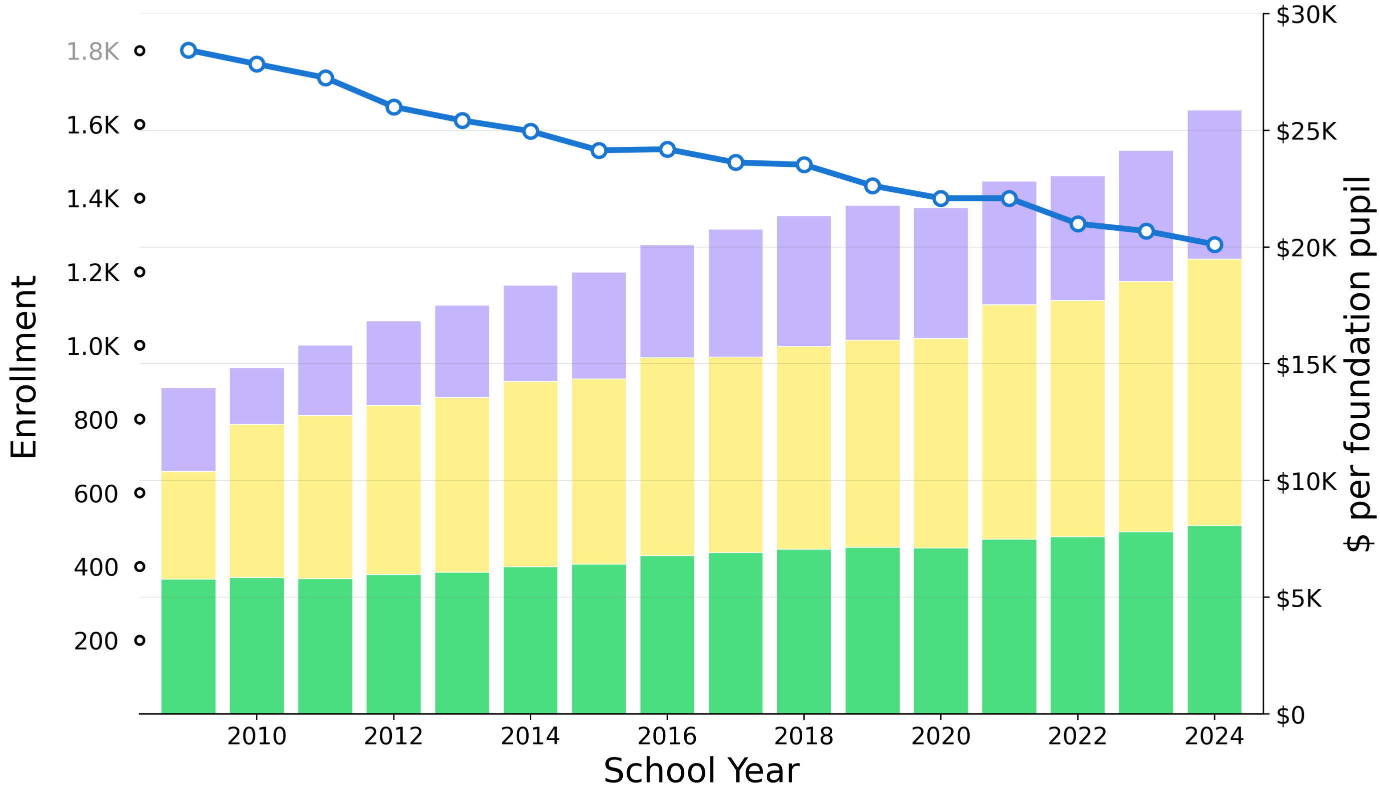


Figure 24

Table 31

	Component	2009 $/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 $/pupil
	Actual NSS (minus Req NSS)	$3,584	3.92%	4.46%	2.07%	$6,383
	Req NSS (minus Ch70)	$4,611	6.23%	3.70%	5.14%	$11,414
	Ch70 Aid	$5,773	2.26%	2.49%	2.49%	$8,068
	Total (Actual NSS)	$13,968	4.19%	3.48%	3.50%	$25,865

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp — Above cohort baseline (Medium (801-1600))

CAGR = (End/Start)^(1/years) – 1 — Below cohort baseline (Medium (801-1600))

→ *Compare to Tiny (0-200 FTE) cohort: Chapter 70 Aid and Net School Spending (page 24)*

Section 3 — Specific districts

Amherst — PPE and Enrollment

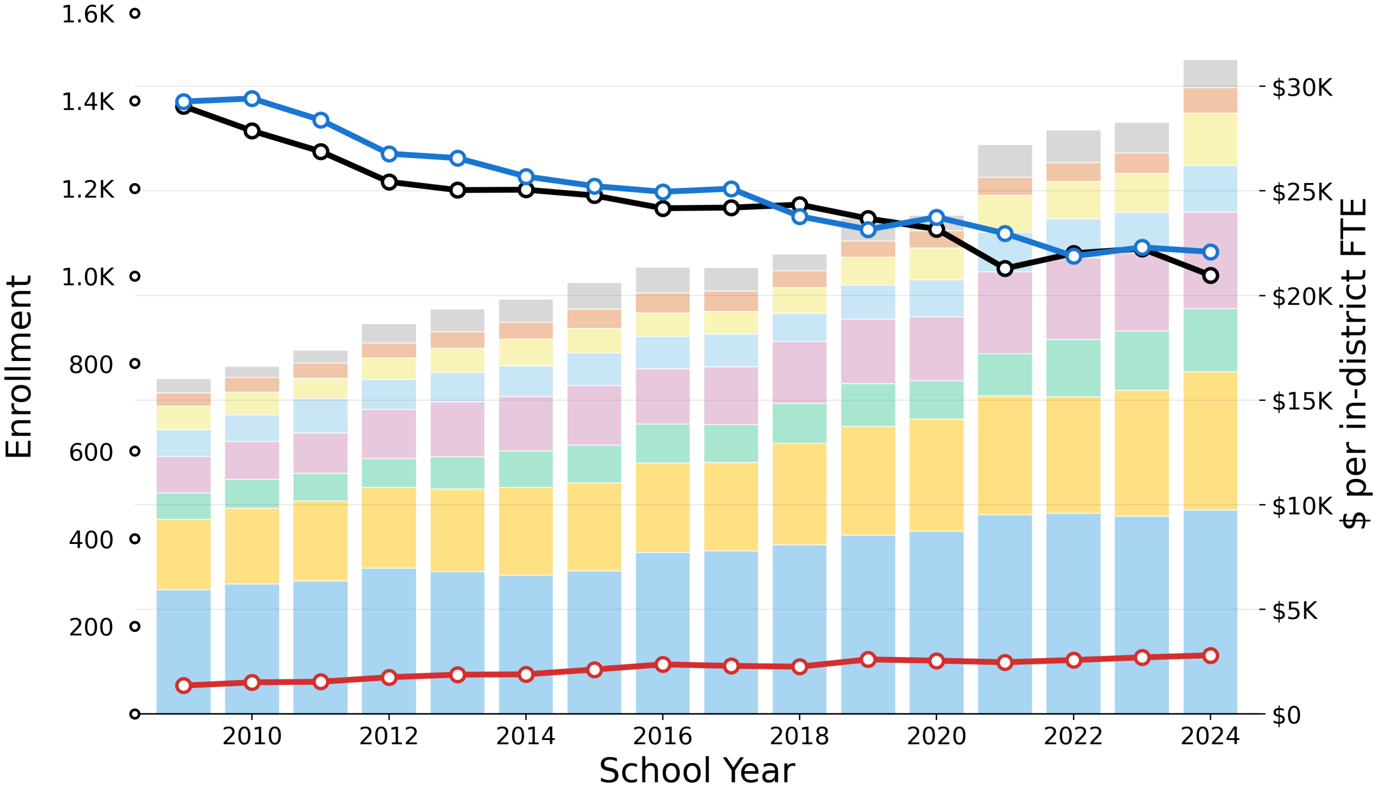


Figure 25

Table 32

	Category	2009 $/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 $/pupil
	Other	$684	+4.5%	+1.9%	+4.0%	$1,328
	Administration	$627	+4.5%	+4.1%	+9.6%	$1,213
	Instructional Leadership	$1,134	+5.4%	+7.0%	+13.5%	$2,498
	Operations and Maintenance	$1,264	+3.9%	+4.5%	+6.4%	$2,248
	Other Teaching Services	$1,762	+6.6%	+5.9%	+8.4%	$4,612
	Pupil Services	$1,261	+6.0%	+5.5%	+8.3%	$3,018
	Insurance, Retirement and Other	$3,353	+4.6%	+4.7%	+4.9%	$6,605
	Teachers	$5,944	+3.4%	+3.9%	+2.7%	$9,745
	Total	$16,029	+4.6%	+4.7%	+5.7%	$31,267

Table 33

	In-district FTE	2009	CAGR 15y	CAGR 10y	CAGR 5y	2024
	In-District FTE Pupils	1,387	-2.1%	-1.8%	-2.4%	1,002
	Foundation Enrollment	1,398	-1.9%	-1.5%	-0.9%	1,055
	Out-of-District FTE Pupils	65	+5.0%	+4.0%	+1.5%	134

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔEnrollment| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp	Above cohort baseline (Medium (801-1600))
CAGR = (End/Start)^(1/years) – 1	Below cohort baseline (Medium (801-1600))

→ *Compare to Medium (801-1600 FTE) cohort: PPE and Enrollment (page 27)*

Section 3 — Selected districts

Amherst — Chapter 70 Aid and Net School Spending (per foundation pupil)

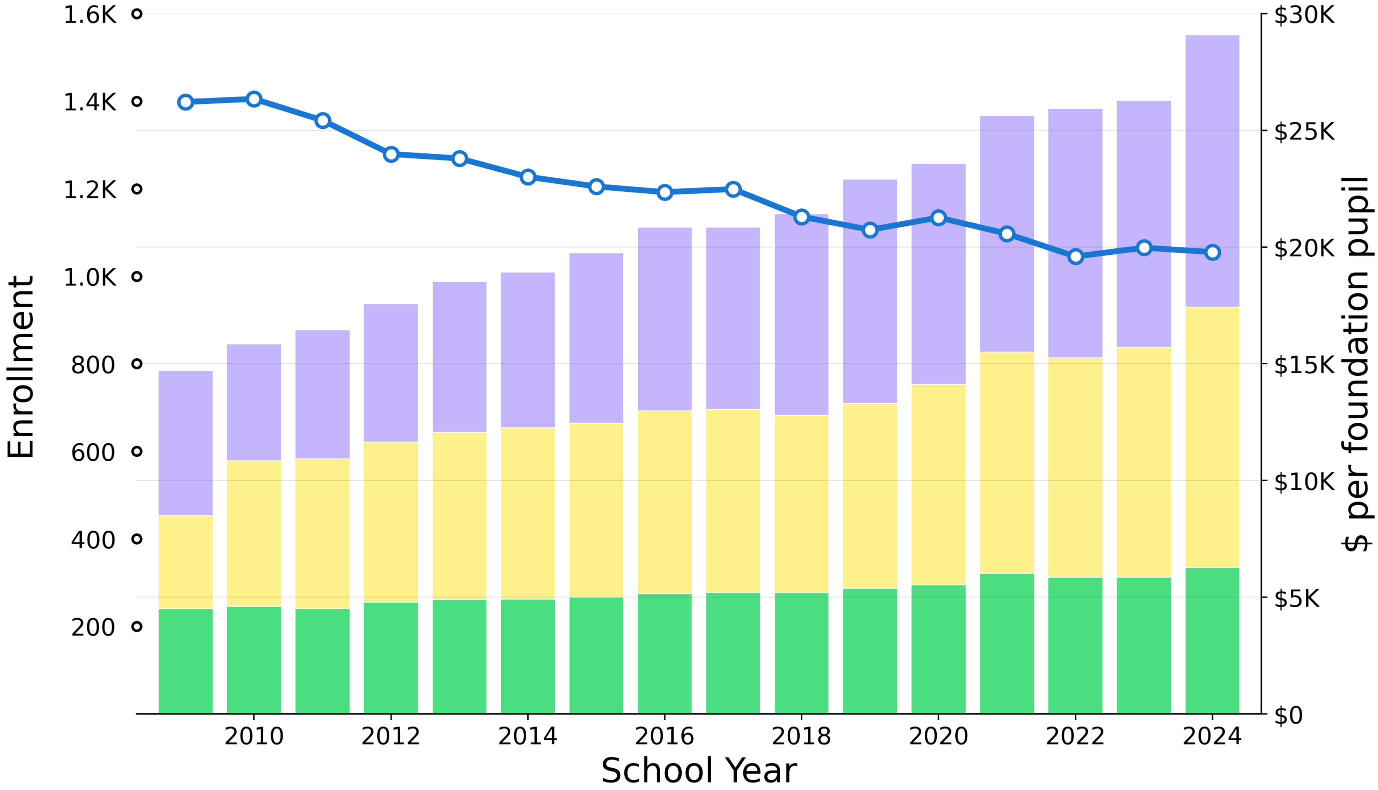


Figure 26

Table 34

	Component	2009 $/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 $/pupil
	Actual NSS (minus Req NSS)	$6,217	4.29%	5.77%	3.94%	$11,670
	Req NSS (minus Ch70)	$3,977	7.11%	4.27%	7.11%	$11,151
	Ch70 Aid	$4,518	2.21%	2.44%	3.10%	$6,271
	Total (Actual NSS)	$14,712	4.65%	4.39%	4.89%	$29,093

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp — Above cohort baseline (Medium (801-1600))

CAGR = (End/Start)^(1/years) – 1 — Below cohort baseline (Medium (801-1600))

→ *Compare to Medium (801-1600 FTE) cohort: Chapter 70 Aid and Net School Spending (page 28)*

Section 3 — Specific districts

Leverett — PPE and Enrollment

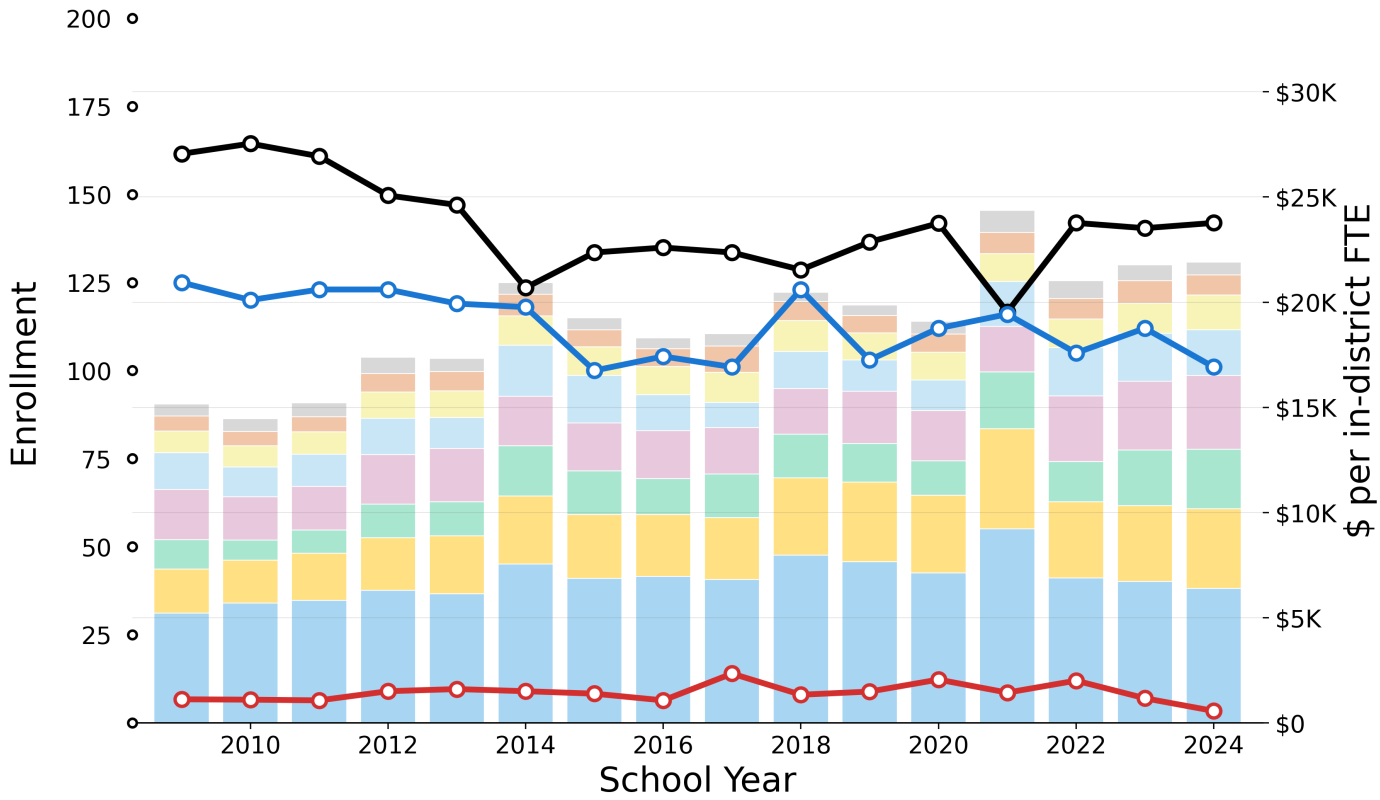


Figure 27

Table 35

	Category	2009 $/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 $/pupil
	Other	$575	+0.4%	+0.9%	+3.9%	$608
	Administration	$715	+1.9%	-0.8%	+2.9%	$953
	Instructional Leadership	$1,015	+3.3%	+1.8%	+5.4%	$1,663
	Operations and Maintenance	$1,755	+1.4%	-1.1%	+7.7%	$2,158
	Other Teaching Services	$2,385	+2.6%	+4.1%	+7.3%	$3,526
	Pupil Services	$1,395	+4.8%	+1.6%	+8.8%	$2,814
	Insurance, Retirement and Other	$2,098	+4.0%	+1.6%	+0.1%	$3,793
	Teachers	$5,218	+1.4%	-1.6%	-3.6%	$6,395
	Total	$15,156	+2.5%	+0.5%	+2.0%	$21,910

Table 36

	In-district FTE	2009	CAGR 15y	CAGR 10y	CAGR 5y	2024
	In-District FTE Pupils	162	-0.9%	+1.4%	+0.8%	142
	Foundation Enrollment	125	-1.4%	-1.5%	-0.4%	101
	Out-of-District FTE Pupils	7	-4.4%	-9.3%	-17.5%	3

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔEnrollment| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp	Above cohort baseline (Tiny (0-200))
CAGR = (End/Start)^(1/years) – 1	Below cohort baseline (Tiny (0-200))

→ *Compare to Tiny (0-200 FTE) cohort: PPE and Enrollment (page 23)*

Section 3 — Selected districts

Leverett — Chapter 70 Aid and Net School Spending (per foundation pupil)

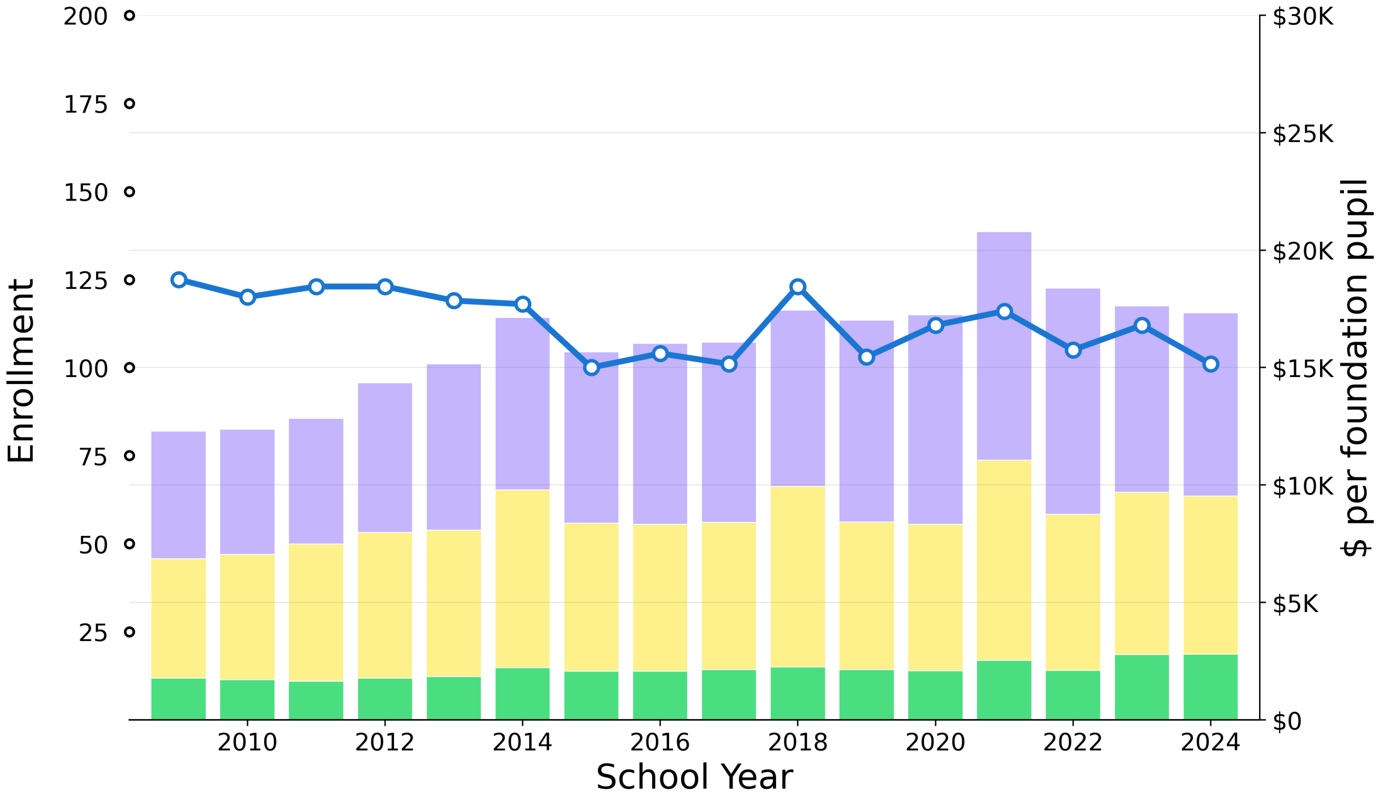


Figure 28 *Table 37*

	Component	2009 $/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 $/pupil
	Actual NSS (minus Req NSS)	$5,429	2.44%	0.62%	-1.89%	$7,799
	Req NSS (minus Ch70)	$5,074	1.90%	-1.17%	1.36%	$6,728
	Ch70 Aid	$1,782	3.05%	2.32%	5.50%	$2,798
	Total (Actual NSS)	$12,285	2.32%	0.12%	0.37%	$17,325

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp — Above cohort baseline (Tiny (0-200))

CAGR = (End/Start)^(1/years) – 1 — Below cohort baseline (Tiny (0-200))

→ *Compare to Tiny (0-200 FTE) cohort: Chapter 70 Aid and Net School Spending (page 24)*

Section 3 — Specific districts

Pelham — PPE and Enrollment

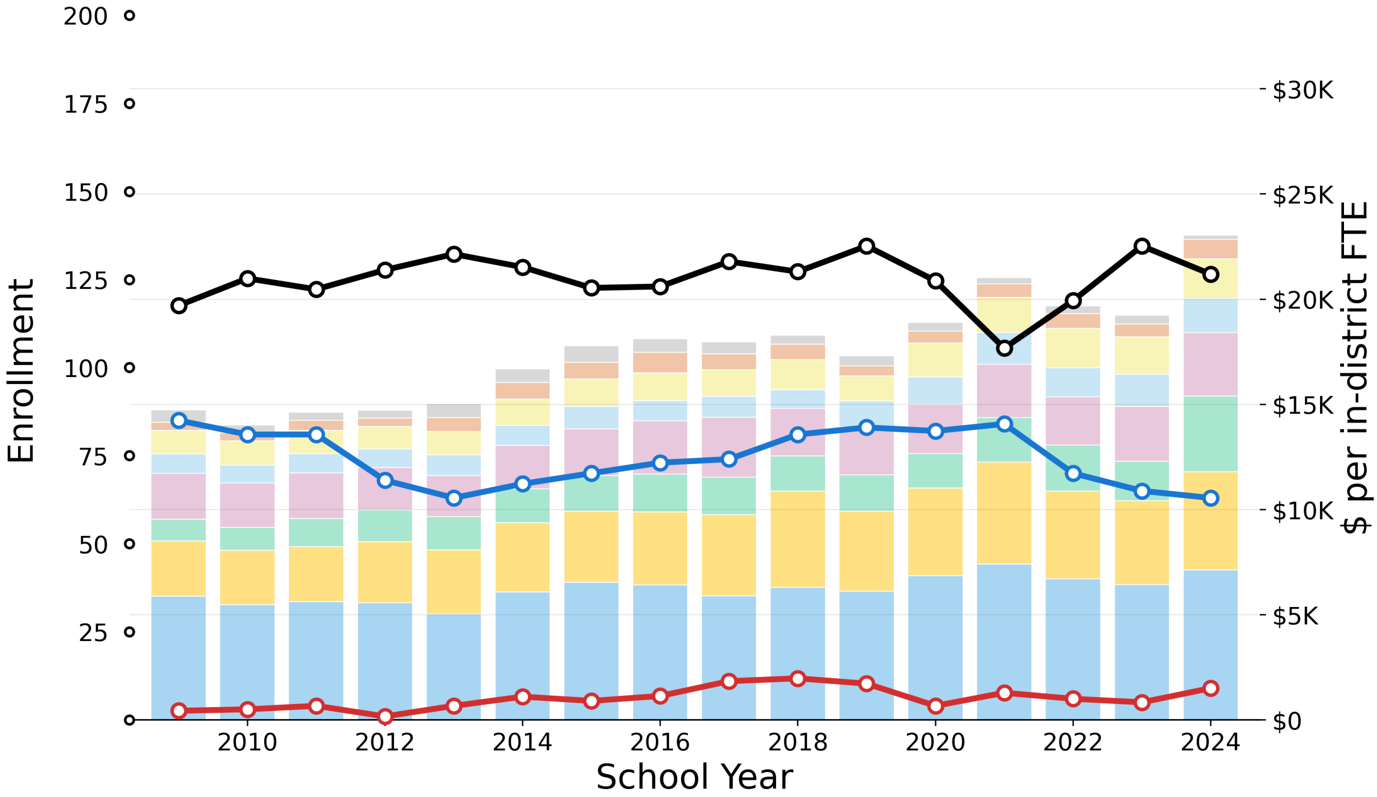


Figure 29

Table 38

	Category	2009 $/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 $/pupil
	Other	$580	-6.6%	-10.7%	-15.0%	$207
	Administration	$390	+5.9%	+1.4%	+13.4%	$917
	Instructional Leadership	$1,123	+3.5%	+4.2%	+9.6%	$1,885
	Operations and Maintenance	$920	+3.8%	+5.5%	+5.6%	$1,614
	Other Teaching Services	$2,184	+2.2%	+3.9%	+5.8%	$3,010
	Pupil Services	$1,032	+8.7%	+8.4%	+15.8%	$3,610
	Insurance, Retirement and Other	$2,615	+3.9%	+3.6%	+4.3%	$4,670
	Teachers	$5,889	+1.3%	+1.6%	+3.1%	$7,127
	Total	$14,733	+3.0%	+3.3%	+5.9%	$23,040

Table 39

In-district FTE	2009	CAGR 15y	CAGR 10y	CAGR 5y	2024
In-District FTE Pupils	118	+0.5%	-0.2%	-1.2%	126
Foundation Enrollment	85	-2.0%	-0.6%	-5.4%	63
Out-of-District FTE Pupils	3	+8.6%	+3.2%	-2.7%	9

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔEnrollment| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp	Above cohort baseline (Tiny (0-200))
CAGR = (End/Start)^(1/years) – 1	Below cohort baseline (Tiny (0-200))

→ *Compare to Tiny (0-200 FTE) cohort: PPE and Enrollment (page 23)*

Section 3 — Selected districts

Pelham — Chapter 70 Aid and Net School Spending (per foundation pupil)

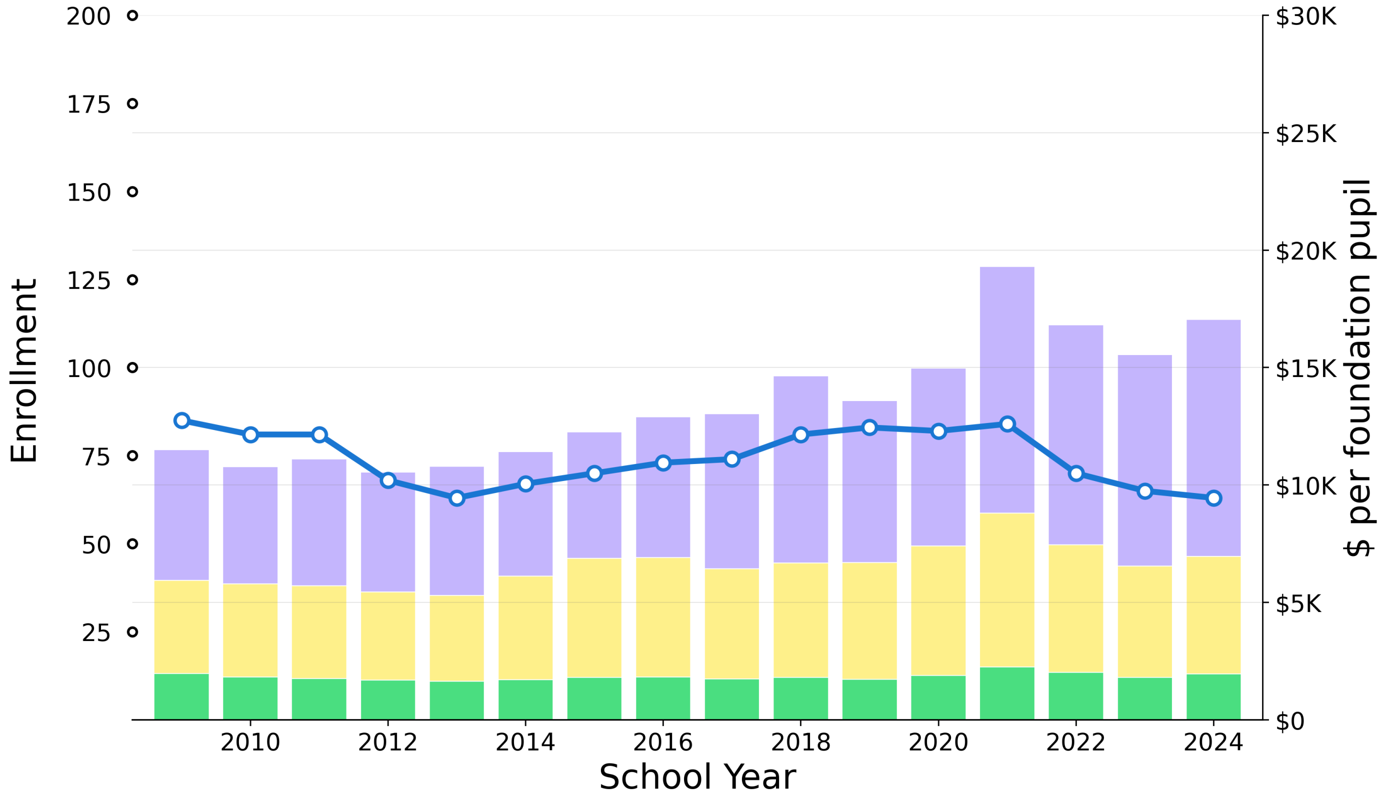


Figure 30 *Table 40*

	Component	2009 $/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 $/pupil
	Actual NSS (minus Req NSS)	$5,561	4.05%	6.64%	7.91%	$10,086
	Req NSS (minus Ch70)	$3,963	1.56%	1.26%	0.10%	$4,996
	Ch70 Aid	$1,983	-0.07%	1.35%	2.52%	$1,963
	Total (Actual NSS)	$11,507	2.65%	4.08%	4.62%	$17,044

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp — Above cohort baseline (Tiny (0-200))

CAGR = (End/Start)^(1/years) – 1 — Below cohort baseline (Tiny (0-200))

→ *Compare to Tiny (0-200 FTE) cohort: Chapter 70 Aid and Net School Spending (page 24)*

Section 3 — Specific districts

Shutesbury — PPE and Enrollment

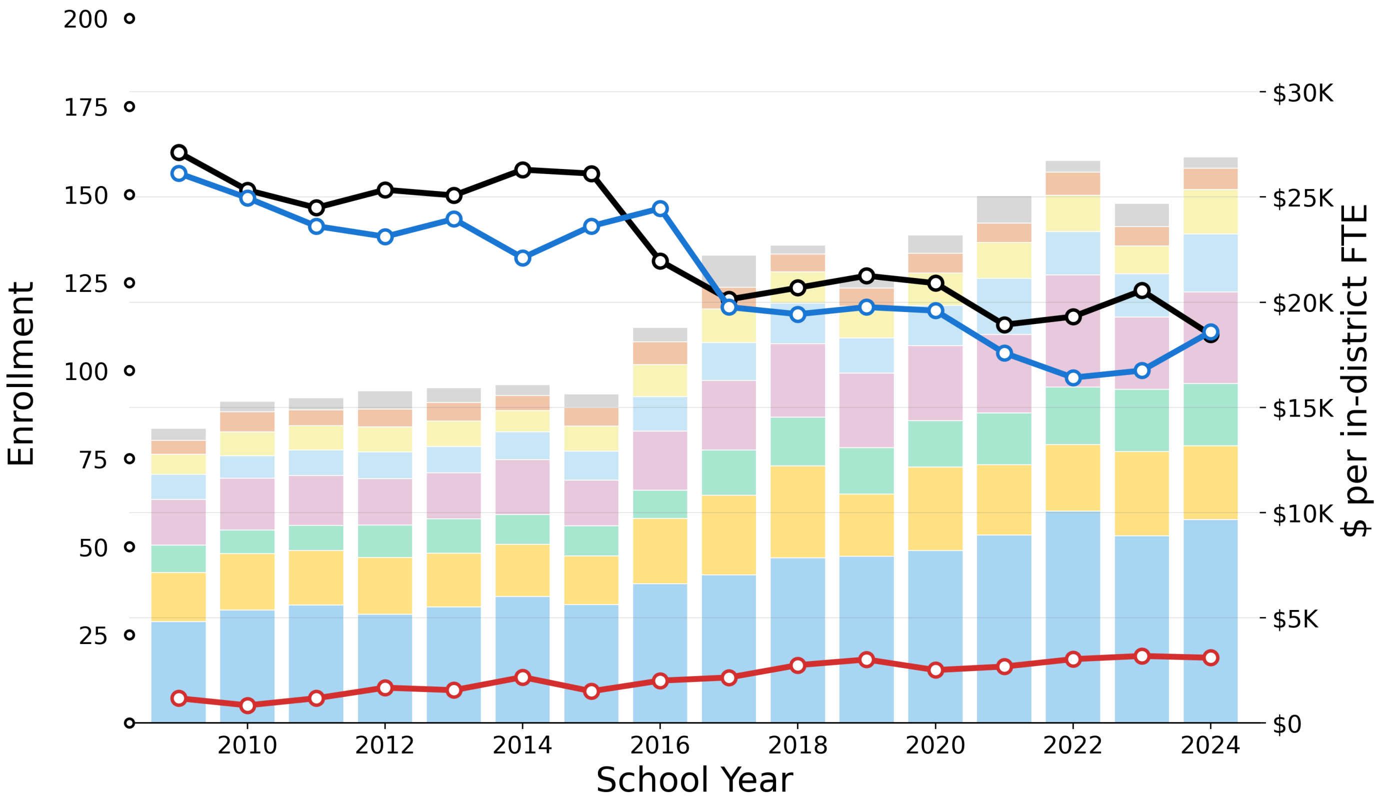


Figure 31

Table 41

	Category	2009 $/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 $/pupil
	Other	$580	-0.4%	+0.6%	+7.2%	$545
	Administration	$652	+3.0%	+3.4%	+3.5%	$1,015
	Instructional Leadership	$969	+5.3%	+7.7%	+6.8%	$2,103
	Operations and Maintenance	$1,180	+5.8%	+7.5%	+10.6%	$2,754
	Other Teaching Services	$2,185	+4.7%	+5.3%	+4.2%	$4,359
	Pupil Services	$1,292	+5.7%	+7.6%	+6.1%	$2,962
	Insurance, Retirement and Other	$2,332	+2.7%	+3.5%	+3.3%	$3,501
	Teachers	$4,821	+4.7%	+4.9%	+4.1%	$9,669
	Total	$14,011	+4.4%	+5.3%	+5.0%	$26,908

Table 42

	In-district FTE	2009	CAGR 15y	CAGR 10y	CAGR 5y	2024
	In-District FTE Pupils	162	-2.5%	-3.5%	-2.8%	110
	Foundation Enrollment	156	-2.2%	-1.7%	-1.2%	111
	Out-of-District FTE Pupils	7	+6.7%	+3.6%	+0.5%	18

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔEnrollment| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp	Above cohort baseline (Tiny (0-200))
CAGR = (End/Start)^(1/years) – 1	Below cohort baseline (Tiny (0-200))

→ *Compare to Tiny (0-200 FTE) cohort: PPE and Enrollment (page 23)*

Section 3 — Selected districts

Shutesbury — Chapter 70 Aid and Net School Spending (per foundation pupil)

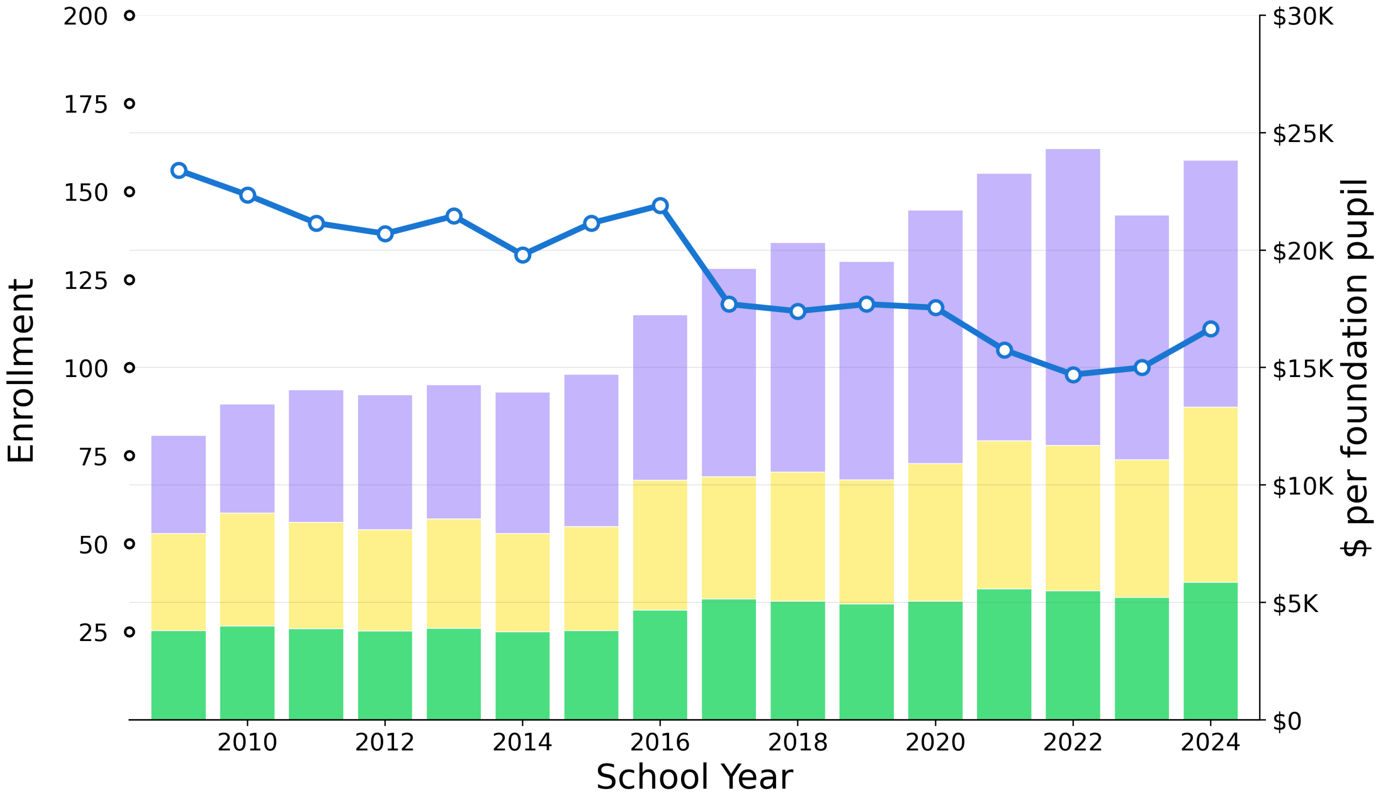


Figure 32

Table 43

	Component	2009 $/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 $/pupil
	Actual NSS (minus Req NSS)	$4,173	6.36%	5.74%	2.51%	$10,526
	Req NSS (minus Ch70)	$4,124	4.02%	5.92%	7.14%	$7,445
	Ch70 Aid	$3,808	2.92%	4.58%	3.48%	$5,862
	Total (Actual NSS)	$12,105	4.62%	5.50%	4.08%	$23,833

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp	Above cohort baseline (Tiny (0-200))
CAGR = (End/Start)^(1/years) – 1	Below cohort baseline (Tiny (0-200))

→ *Compare to Tiny (0-200 FTE) cohort: Chapter 70 Aid and Net School Spending (page 24)*